



Analytic Number Theory Note

Introduction to the Theory of Numbers

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Chapter 1 Introduction

1.1 Basic Definitions

Definition 1.1 (Natural Numbers)

The set of natural numbers is denoted by $\mathbb{N} = \{1, 2, 3, \dots\}$.



Definition 1.2 (Prime Numbers)

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself. And the set of prime numbers is denoted by $\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$.



Definition 1.3 (Prime Counting Function)

We using the notation $\pi(N) = \#\{p \leq N : p \in \mathbb{P}\}$ to denote the number of primes less than or equal to N . Sometimes, we also use $\pi(N) = \sum_{p \leq N} 1$ to denote the same function.



Definition 1.4 (Euler Totient Function)

For any natural number n , the Euler totient function $\phi(n)$ is defined as the number of integers from 1 to n that are coprime to n . In other words,

$$\phi(n) = \#\{k \in \mathbb{N} : 1 \leq k \leq n, \gcd(k, n) = 1\}.$$



1.2 Fundamental Theorem of Arithmetic and Infinitely Many Primes

Theorem 1.1 (Fundamental Theorem of Arithmetic (FTA))

Every natural number greater than 1 can be uniquely factored into prime numbers, up to the order of the factors. In other words, for any $n \in \mathbb{N}$ with $n > 1$, there exist unique primes $p_1 < p_2 < \dots < p_k$ and positive integers $\alpha_1, \alpha_2, \dots, \alpha_k$ such that

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}.$$



Proof First, we show existence. We proceed by induction on n . The base case $n = 2$ is prime itself. Assume true for all integers less than n . If n is prime, we are done. If not, let $n = ab$ with $1 < a, b < n$. By the induction hypothesis, both a and b can be factored into primes. Combining these factorizations gives a prime factorization of n .

Next, we show uniqueness. Suppose n has two distinct prime factorizations:

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k} = q_1^{\beta_1} q_2^{\beta_2} \dots q_m^{\beta_m},$$

where p_i and q_j are primes. Without loss of generality, let p_1 be the smallest prime in the first factorization. Since p_1 divides n , it must divide the right-hand side. By the property of primes, p_1 must equal one of the q_j . Repeating this argument for all primes in both factorizations shows that the multisets of primes must be identical, establishing uniqueness.

Theorem 1.2 (Infinitely Many Primes)

Over 2,300 years ago, Euclid showed that there are infinitely many primes. (i.e. $\lim_{N \rightarrow \infty} \pi(N) = \infty$)



Proof Assume, for the sake of contradiction, that there are only finitely many primes: $p_1, p_2, \dots, p_k$. Consider the number

$$N = p_1 p_2 \dots p_k + 1.$$

By construction, N is not divisible by any of the primes $p_1, p_2, \dots, p_k$, since it leaves a remainder of 1 when divided

by any of them. Therefore, either N is prime itself or it has prime factors not in our original list. In either case, this contradicts our assumption that we had listed all primes. Hence, there must be infinitely many primes.

Proposition 1.1 (Lower Bound for $\pi(n)$)

For any natural number $n \geq 1$, we have the following inequality for $\pi(n)$:

$$\log \log n \leq \pi(n)$$

Exercise 1.1 Prove the Proposition 1.1. (workshop 1)

(Hint: Let p_n be the n -th prime number. Show that $p_n \leq 2^{2^{n-1}}$ for all $n \geq 1$.)

Solution First, we prove by induction that $p_n \leq 2^{2^{n-1}}$ for all $n \geq 1$.

The base case $n = 1$ holds since $p_1 = 2 \leq 2^{2^0} = 2$. Assume the statement is true for some $k \geq 1$, i.e., $p_k \leq 2^{2^{k-1}}$. We need to show it holds for $k + 1$. Consider the number

$$N = p_1 p_2 \cdots p_k + 1.$$

By construction, N is not divisible by any of the primes $p_1, p_2, \dots, p_k$. Therefore, either N is prime itself or it has a prime factor greater than p_k . In either case, we have

$$p_{k+1} \leq N = p_1 p_2 \cdots p_k + 1 \leq (2^{2^0})(2^{2^1}) \cdots (2^{2^{k-1}}) + 1 = 2^{\sum_{i=0}^{k-1} 2^i} + 1 = 2^{2^k - 1} + 1 \leq 2^{2^k}.$$

Thus, by induction, the inequality $p_n \leq 2^{2^{n-1}}$ holds for all $n \geq 1$.

Now, to prove the proposition, let $n \geq 1$ and let $k = \pi(n)$, which is the number of primes less than or equal to n . By the definition of $\pi(n)$, we have $p_k \leq n < p_{k+1}$. Using the inequality we just proved, we get

$$n < p_{k+1} \leq 2^{2^k} \implies \log n \leq 2^k \log 2 \implies \log \log n \leq k \log 2 + \log \log 2 \leq k = \pi(n).$$

Therefore, we conclude that $\log \log n \leq \pi(n)$ for all $n \geq 1$.

Theorem 1.3 (Prime Number Theorem (PNT))

As N approaches infinity, the prime counting function $\pi(N)$ is asymptotically equivalent to $\frac{N}{\log N}$: $\pi(N) \sim \frac{N}{\log N}$ as $N \rightarrow \infty$. In other words,

$$\lim_{N \rightarrow \infty} \frac{\pi(N)}{\frac{N}{\log N}} = 1.$$

The proof of Theorem 1.3 is one of the main goal of this course and will be given in later chapters after introducing more tools from complex analysis.

Lemma 1.1

To prove Theorem 1.3, it equivalent to show that for any positive constants c_1, c_2 with $c_1 < 1 < c_2$, we have

$$c_1 \frac{N}{\log N} \leq \pi(N) \leq c_2 \frac{N}{\log N}$$

Proof Suppose Theorem 1.3 holds. Then for any $\varepsilon > 0$, there exists N_0 such that for all $N > N_0$,

$$\left| \frac{\pi(N)}{\frac{N}{\log N}} - 1 \right| < \varepsilon.$$

Choosing $\varepsilon = \min(1 - c_1, c_2 - 1)$, we have

$$1 - \varepsilon < \frac{\pi(N)}{\frac{N}{\log N}} < 1 + \varepsilon,$$

which implies

$$c_1 \frac{N}{\log N} < \pi(N) < c_2 \frac{N}{\log N}.$$

Conversely, suppose the inequalities hold for all sufficiently large N . Then for any $\varepsilon > 0$, we can choose $c_1 = 1 - \varepsilon$

and $c_2 = 1 + \varepsilon$. Thus, for all sufficiently large N ,

$$(1 - \varepsilon) \frac{N}{\log N} < \pi(N) < (1 + \varepsilon) \frac{N}{\log N},$$

which leads to

$$-\varepsilon < \frac{\pi(N)}{\frac{N}{\log N}} - 1 < \varepsilon.$$

Therefore,

$$\left| \frac{\pi(N)}{\frac{N}{\log N}} - 1 \right| < \varepsilon,$$

proving Theorem 1.3.

Remark A more precise version of Theorem 1.3 states that

$$\pi(x) = \text{Li}(x) + E(x) \tag{1.1}$$

where $\text{Li}(x) = \int_2^x \frac{dt}{\log t} \sim \frac{x}{\log x}$ and the error term $E(x)$ satisfies $E(x) = \mathcal{O}(\sqrt{x} \log x)$.

 **Exercise 1.2** Prove that equation (1.1) implies Theorem 1.3.

Solution From equation (1.1), we have

$$\pi(x) = \text{Li}(x) + E(x).$$

Dividing both sides by $\frac{x}{\log x}$, we get

$$\frac{\pi(x)}{\frac{x}{\log x}} = \frac{\text{Li}(x)}{\frac{x}{\log x}} + \frac{E(x)}{\frac{x}{\log x}}.$$

As $x \rightarrow \infty$, we know that $\text{Li}(x) \sim \frac{x}{\log x}$, so

$$\lim_{x \rightarrow \infty} \frac{\text{Li}(x)}{\frac{x}{\log x}} = 1.$$

Next, we analyze the error term:

$$\left| \frac{E(x)}{\frac{x}{\log x}} \right| = \left| E(x) \cdot \frac{\log x}{x} \right| = \mathcal{O} \left(\sqrt{x} \log x \cdot \frac{\log x}{x} \right) = \mathcal{O} \left(\frac{(\log x)^2}{\sqrt{x}} \right).$$

Since $\frac{(\log x)^2}{\sqrt{x}} \rightarrow 0$ as $x \rightarrow \infty$, we have

$$\lim_{x \rightarrow \infty} \frac{E(x)}{\frac{x}{\log x}} = 0.$$

Combining these results, we find

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\frac{x}{\log x}} = 1 + 0 = 1.$$

Thus, equation (1.1) implies Theorem 1.3.

Definition 1.5 (Riemann Zeta Function (Outline))

The Riemann zeta function $\zeta(z)$ is defined for complex numbers z and for $\Re(z) > 1$ is given by the infinite series

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}.$$

for the other values of z , $\zeta(z)$ is defined by analytic continuation as it has an analytic continuation to the whole complex plane except $z = 1$.

In the following chapters, we will show that

$$\zeta(1-z) = 2^{1-z} \pi^{-z} \cos\left(\frac{\pi z}{2}\right) \Gamma(z) \zeta(z) \quad \text{for all } z \in \mathbb{C}.$$

and thus $\zeta(-2n) = 0$ which are called the trivial zeros of $\zeta(z)$.



Moreover, if we let $\{\rho\}$ be the set of non-trivial zeros of $\zeta(z)$, then we have the following explicit formula for $\pi(x)$:

$$\pi(x) = \text{Li}(x) - \sum_{\rho} \text{Li}(x^{\rho}) + \text{smaller order terms}.$$

Proposition 1.2 (Analytic Continuation of Zeta Function)

The Riemann zeta function $\zeta(z)$ is holomorphic for all z with $\Re(z) > 1$.

 **Exercise 1.3** Show the proposition 1.2 by Weierstrass M-test.

Solution To show that the Riemann zeta function $\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$ is holomorphic for all z with $\Re(z) > 1$, we will use the Weierstrass M-test.

Let $z = x + iy$ where $x = \Re(z)$ and $y = \Im(z)$. For $\Re(z) > 1$, we have $x > 1$. We can write

$$\left| \frac{1}{n^z} \right| = \frac{1}{|n^z|} = \frac{1}{n^{\Re(z)}} = \frac{1}{n^x}.$$

Since $x > 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n^x}$ converges.

Now, we can choose $M_n = \frac{1}{n^x}$ as our bounding sequence. Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^x}$ converges for $x > 1$, by the Weierstrass M-test, the series $\sum_{n=1}^{\infty} \frac{1}{n^z}$ converges uniformly on any compact subset of the half-plane $\Re(z) > 1$.

Since each term $\frac{1}{n^z}$ is holomorphic in z for $\Re(z) > 1$, and the series converges uniformly, it follows that the sum $\zeta(z)$ is also holomorphic in this region.

Therefore, we conclude that the Riemann zeta function $\zeta(z)$ is holomorphic for all z with $\Re(z) > 1$.

Postulate 1.1 (Riemann Hypothesis (RH))

All non-trivial zeros ρ of the Riemann zeta function $\zeta(z)$ have real part equal to $\frac{1}{2}$, i.e., $\Re(\rho) = \frac{1}{2}$.

Remark One can find that the non-trivial zeros of $\zeta(z)$ are only appear in the critical strip: $0 < \Re(z) < 1$.

Definition 1.6

Usually, we using the notation $(RH)_{\sigma}$ to denote the following statement:

$$\zeta(\rho) \neq 0 \quad \text{for all } \rho \text{ with } \Re(\rho) > \sigma.$$

where ρ is the set of non-trivial zeros of $\zeta(z)$.

By extend Euler Product Formula 1.4 to complex variable, one can show that $\zeta(z) \neq 0$ for all z with $\Re(z) > 1$. Hence, $(RH)_1$ is true. And the Riemann Hypothesis 1.1 is equivalent to $(RH)_{\frac{1}{2}}$.

A fact is that for $\sigma < 1$, $(RH)_{\sigma}$ is equivalent to the following statement about the error term $E(x)$ in equation (1.1):

$$|\pi(x) - \text{Li}(x)| = \mathcal{O}(x^{\sigma} \log x).$$

and by using this fact, it can be shown the Prime Number Theorem 1.3. But unfortunately, we have no ideal if $(RH)_{\sigma}$ is true for any $\sigma < 1$.

However, Hadamard and de la Vallée Poussin (1896) independently proved that by using the fact that $(RH)_1$ is true (i.e. $\zeta(z) \neq 0$ for $\Re(z) > 1$), one can prove the Prime Number Theorem 1.3.

 **Note** Why the zeros of $\zeta(z)$ is important for the distribution of primes?

From the Complex Analysis course, we know that a holomorphic function is completely determined by its zeros and poles (i.e. Singularities). In order to using complex analysis to study $\zeta(z)$, one introduces a fixed function which has the same zeros and poles as $\zeta(z)$, called the Riemann Xi function $\xi(z)$:

$$\xi(z) = \frac{1}{2} z(z-1) \pi^{-\frac{z}{2}} \Gamma\left(\frac{z}{2}\right) \zeta(z).$$

where $\Gamma(z)$ is the Gamma function.

A famous result about $\xi(z)$ is the Hadamard Factorization Theorem:

$$\xi(z) = e^{A+Bz} \prod_{\rho} \left(1 - \frac{z}{\rho}\right) e^{\frac{z}{\rho}}$$

where the product is over all non-trivial zeros ρ of $\zeta(z)$, and A and B are constants.

So once we know the distribution of zeros of $\zeta(z)$, we can understand the behavior of $\xi(z)$ and hence $\zeta(z)$, which in turn gives us information about the distribution of prime numbers via the explicit formula for $\pi(x)$.

1.3 Euler Product Formula and Dirichlet Theorem

Theorem 1.4 (Euler Product Formula)

Thinking the zeta function as a function of real variable $x > 1$, we have the Euler Product Formula:

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-x}} \quad \text{for } x > 1.$$



Proof Recall the geometric series formula: for $|r| < 1$, we have

$$\frac{1}{1-r} = \sum_{k=0}^{\infty} r^k = 1 + r + r^2 + r^3 + \dots$$

Setting $r = p^{-x} < 1$ for each prime p , we get

$$\frac{1}{1-p^{-x}} = \sum_{k=0}^{\infty} p^{-kx} = 1 + p^{-x} + p^{-2x} + p^{-3x} + \dots$$

Consider two distinct primes p and q . Multiplying their corresponding series, we have

$$(1 + p^{-x} + p^{-2x} + \dots)(1 + q^{-x} + q^{-2x} + \dots) = \sum_{j,k \geq 0} \frac{1}{[p^j q^k]^x}$$

and so by the Fundamental Theorem of Arithmetic 1.1, the number of the form $p^j q^k$ are precisely the number n whose prime factors are only p and q . Hence

$$\frac{1}{1-p^{-x}} \cdot \frac{1}{1-q^{-x}} = \sum_{n \in N_{p,q}} \frac{1}{n^x} \quad \text{where } N_{p,q} = \{n \in \mathbb{N} : n = p^j q^k, j, k \in \mathbb{N}_0\}.$$

Extending this argument to all primes less than or equal to integer N , let $E_N = \{n \in \mathbb{N} : \text{all prime factors of } n \text{ are } \leq N\}$, we have

$$\prod_{p \leq N} \frac{1}{1-p^{-x}} = \sum_{n \in E_N} \frac{1}{n^x}.$$

Now, we show that as $N \rightarrow \infty$, we can get Euler Product Formula. Let

$$S_N = \sum_{n \in E_N} \frac{1}{n^x} \quad T_N = \zeta(x) - S_N = \sum_{n \notin E_N} \frac{1}{n^x}.$$

For any $n \notin E_N$, n must have a prime factor greater than N . Thus, $n = pm$ for some prime $p > N$ and some $m \in \mathbb{N}$. Therefore,

$$T_N \leq \sum_{\substack{p > N \\ p \in \mathbb{P}}} \sum_{m=1}^{\infty} \frac{1}{(pm)^x} = \sum_{\substack{p > N \\ p \in \mathbb{P}}} \frac{1}{p^x} \sum_{m=1}^{\infty} \frac{1}{m^x} = \zeta(x) \sum_{\substack{p > N \\ p \in \mathbb{P}}} \frac{1}{p^x}.$$

Since the series $\sum_{p \in \mathbb{P}} \frac{1}{p^x}$ converges for $x > 1$ by p-series test and comparison test, the tail $\sum_{p > N} \frac{1}{p^x} \rightarrow 0$ as $N \rightarrow \infty$. Hence, $T_N \rightarrow 0$ as $N \rightarrow \infty$, which implies that $S_N \rightarrow \zeta(x)$. Thus we have

$$\zeta(x) = \lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \prod_{\substack{p \leq N \\ p \in \mathbb{P}}} \frac{1}{1-p^{-x}} = \prod_{p \in \mathbb{P}} \frac{1}{1-p^{-x}}.$$

Corollary 1.1 (Divergence of the Sum of Reciprocals of Primes)

By taking the logarithmic derivative of the Euler Product Formula, we can get the following fact

$$\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty.$$



Proof Taking the logarithm of both sides of the Euler Product Formula, we have

$$\log \zeta(x) = - \sum_{p \in \mathbb{P}} \log(1 - p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{p^{-kx}}{k}$$

where the last equality follows from the Taylor series expansion of $\log(1 - y) = -\sum_{k=1}^{\infty} \frac{y^k}{k}$ for $0 \leq y < 1$.

For the inner sum, we have

$$\sum_{k=2}^{\infty} \frac{p^{-kx}}{k} < \sum_{k=2}^{\infty} p^{-k} = \frac{p^{-2}}{1 - p^{-1}} = \frac{1}{p(p-1)}.$$

Thus,

$$\log \zeta(x) = \sum_{p \in \mathbb{P}} \left(\frac{p^{-x}}{1} + \sum_{k=2}^{\infty} \frac{p^{-kx}}{k} \right) = \sum_{p \in \mathbb{P}} p^{-x} + \sum_{p \in \mathbb{P}} \sum_{k=2}^{\infty} \frac{p^{-kx}}{k} \leq \sum_{p \in \mathbb{P}} p^{-x} + E(x)$$

where $E(x) = \sum_{p \in \mathbb{P}} \frac{1}{p(p-1)}$ is a convergent series, and

$$E(x) \leq \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1.$$

Therefore,

$$\log \zeta(x) \leq \sum_{p \in \mathbb{P}} p^{-x} + 1.$$

As $x \rightarrow 1^+$, $\zeta(x) \rightarrow \infty$, so $\log \zeta(x) \rightarrow \infty$ by continuity of logarithm. Hence,

$$\sum_{p \in \mathbb{P}} p^{-x} \rightarrow \infty \quad \text{as } x \rightarrow 1^+.$$

and thus,

$$\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty.$$

Definition 1.7 (Arithmetic Progression (AP))

An arithmetic progression is a sequence of numbers in which the difference between consecutive terms is constant. It can be expressed in the form:

$$\text{AP}_{a,q} = \{a + q, a + 2q, a + 3q, \dots\} = \{nq + a : n \in \mathbb{N}\}$$

where a is the first term and q is the common difference.



Proposition 1.3 (Finiteness of Primes in Certain APs)

If $d = \gcd(a, q) > 1$, then the arithmetic progression $\text{AP}_{a,q} = \{nq + a : n \in \mathbb{N}\}$ contains only finitely many primes.



Exercise 1.4 Prove the proposition above.

Solution If $d = \gcd(a, q) > 1$, then every term in the arithmetic progression can be expressed as

$$nq + a = n(dk) + (dm) = d(nk + m)$$

for some integers k and m . This means that every term in the progression is divisible by d . Since $d > 1$, none of these terms can be prime (except possibly the first term if it equals d itself). Therefore, the arithmetic progression contains only finitely many primes.

Theorem 1.5 (Dirichlet's Theorem on Arithmetic Progressions)

If a and q are coprime (i.e., $\gcd(a, q) = 1$), then the arithmetic progression $\text{AP}_{a,q} = \{nq + a : n \in \mathbb{N}\}$ contains infinitely many primes.



The proof of Dirichlet's Theorem requires more advanced tools from analytic number theory, including Dirichlet characters and L-functions, which will be introduced in later chapters. However, various special cases and consequences of this theorem can be explored with more elementary methods. The following exercise provides an opportunity to investigate one such special case.

Exercise 1.5 Give an elementary proof of Dirichlet's Theorem for the AP $\{4n - 1 : n \in \mathbb{N}\}$ by completing the following

outline:

- Argue by contradiction as Euclid did for the infinitude of primes and suppose there are only finitely many primes of the form $4n - 1$, say $p_1, p_2, \dots, p_k$.
- Consider the number $N := 2^2 \cdot 3 \cdot 5 \cdots p_k - 1$. (all odd primes up to p_k).
- Note that $N > p$ and $N = 4m - 1$ for some $m \in \mathbb{N}$.
- Hence N is not prime. Deduce that N has a prime factor of the form $4n - 1$ to obtain a contradiction.

Solution We proceed by contradiction. Suppose there are only finitely many primes of the form $4n - 1$, say $p_1, p_2, \dots, p_k$. Consider the number

$$N := 2^2 \cdot 3 \cdot 5 \cdots p_k - 1.$$

Note that $N > p_k$ since $p_k \geq 3$. Also, we can express N in the form $4m - 1$ for some integer m .

Since N is greater than any of the primes $p_1, p_2, \dots, p_k$, it cannot be prime itself (as it would contradict our assumption). Therefore, N must be composite and have prime factors.

Now, we analyze the prime factors of N . Any prime factor of N must be either of the form $4n - 1$ or $4n + 1$. (This is because all primes greater than 2 are odd, and odd primes can be expressed in one of these two forms.) However, if all prime factors were of the form $4n + 1$, then their product would also be of the form $4n + 1$. This is because the product of two numbers of the form $4n + 1$ is also of that form:

$$(4a + 1)(4b + 1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1.$$

Since we have established that N is of the form $4m - 1$, it cannot be expressed as a product of only primes of the form $4n + 1$. Therefore, at least one prime factor of N must be of the form $4n - 1$. But this contradicts our initial assumption that $p_1, p_2, \dots, p_k$ were the only primes of that form and all of them not dividing N .

Hence, we conclude that there are infinitely many primes of the form $4n - 1$.

Lemma 1.2

Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\text{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \pmod{q}, \\ 0 & \text{otherwise.} \end{cases}$$

Then once we have

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

we can show that the series

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p} = \infty$$

and hence Dirichlet's Theorem 1.5 holds.



Proof Dirichlet was inspired by Euler's proof that the series of reciprocals of primes diverges (Corollary 1.1) to prove his theorem. Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\text{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \pmod{q}, \\ 0 & \text{otherwise.} \end{cases}$$

Then once can show that the series

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p} = \infty$$

for any a, q with $\gcd(a, q) = 1$, which implies that there are infinitely many primes in the arithmetic progression $\text{AP}_{a,q}$ and hence Dirichlet's Theorem 1.5 holds. Then one might hope extend Euler's Product Formula 1.4 to

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

On the other hand, since

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n} = \sum_{m=1}^{\infty} \frac{1}{mq+a} \geq \frac{1}{a+q} \sum_{m=1}^{\infty} \frac{1}{m} = \infty.$$

we see that $L(x; \mathbb{1}_{a,q}) := \sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x}$ tends to infinity as $x \rightarrow 1^+$. Thus, do the similar things as in Corollary 1.1, we have

$$\log L(x; \mathbb{1}_{a,q}) = - \sum_{p \in \mathbb{P}} \log(1 - \mathbb{1}_{a,q}(p)p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k}.$$

where the last equality follows from the Taylor series expansion of $\log(1-y) = -\sum_{k=1}^{\infty} \frac{y^k}{k}$ for $0 \leq y < 1$. And as before, since

$$\sum_{k=2}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k} \leq \sum_{k=2}^{\infty} p^{-k} = \frac{p^{-2}}{1-p^{-1}} = \frac{1}{p(p-1)},$$

we have

$$\log L(x; \mathbb{1}_{a,q}) = \sum_{p \in \mathbb{P}} \left(\frac{\mathbb{1}_{a,q}(p)p^{-x}}{1} + \sum_{k=2}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k} \right) = \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} + E(x) \quad (1.2)$$

where $E(x) = \sum_{p \in \mathbb{P}} \frac{1}{p(p-1)} \leq 1$ is a convergent series as before. Therefore,

$$\log L(x; \mathbb{1}_{a,q}) \leq \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} + 1.$$

As $x \rightarrow 1^+$, $L(x; \mathbb{1}_{a,q}) \rightarrow \infty$, so $\log L(x; \mathbb{1}_{a,q}) \rightarrow \infty$ by continuity of logarithm. Hence,

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} \rightarrow \infty \quad \text{as } x \rightarrow 1^+.$$

and thus,

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p} = \infty.$$



Note Note that if we want to show

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

in the above Lemma 1.2. Examining the proof of Euler Product Formula 1.4, we need to show that $\mathbb{1}_{a,q}(n)$ is completely multiplicative. (i.e. $\mathbb{1}_{a,q}(p^j q^k) = \mathbb{1}_{a,q}(p)^j \mathbb{1}_{a,q}(q)^k$). But unfortunately, $\mathbb{1}_{a,q}(n)$ is not completely multiplicative on arithmetic progression. So Lemma 1.2 just gives the idea of Dirichlet's proof of Theorem 1.5, but we need more advanced tools to complete the proof.

One way to fix this problem is to introduce Dirichlet characters χ modulo q , which will be discussed in later chapters. The core idea is to show the formula

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-x}} \quad \text{for } x > 1.$$

and then using the similar argument as in Lemma 1.2 but replacing $\mathbb{1}_{a,q}(n)$ by Dirichlet characters χ modulo q : $\mathbb{1}_{a,q}(n) = \sum_{\chi} c_{\chi} \chi(n)$ (which will be discussed in later chapters) to get

$$\log L(x; \chi) = - \sum_{p \in \mathbb{P}} \log(1 - \chi(p)p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{\chi(p)^k p^{-kx}}{k}.$$

as before, where $L(x; \chi) := \sum_{n=1}^{\infty} \frac{\chi(n)}{n^x}$ and

$$\sum_{\chi} c_{\chi} \log L(x; \chi) = \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} + E(x)$$

as in equation (1.2). At last, we just need $\sum_{\chi} c_{\chi} \log L(x; \chi) \rightarrow \infty$ as $x \rightarrow 1^+$, the following Lemma 1.3 give us $\log L(x; \chi_0) \rightarrow \infty$, and later chapters will show that:

- $L(x; \chi)$ is bounded for all $\chi \neq \chi_0$ as $x \rightarrow 1^+$.
- $L(x; \chi) \neq 0$ for all $\chi \neq \chi_0$ as $x \rightarrow 1^+$.

by using these two properties, we can show that $\sum_{\chi} c_{\chi} \log L(x; \chi) \rightarrow \infty$ as $x \rightarrow 1^+$. Finally, we arrive at the conclusion of Dirichlet's Theorem 1.5.

Lemma 1.3

For principal Dirichlet character χ_0 modulo q

$$\chi_0(n) = \begin{cases} 1 & \text{if } \gcd(n, q) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

we have

$$L(x; \chi_0) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^x} \rightarrow \infty \quad \text{as } x \rightarrow 1^+.$$



Proof By the definition of principal Dirichlet character χ_0 modulo q , we have

$$L(x; \chi_0) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^x} = \sum_{m=0}^{\infty} \sum_{\substack{b=1 \\ \gcd(b, q)=1}}^q \frac{1}{(mq+b)^x} \geq \sum_{m=0}^{\infty} \sum_{\substack{b=1 \\ \gcd(b, q)=1}}^q \frac{1}{(mq+q)^x} = \phi(q) \sum_{m=1}^{\infty} \frac{1}{(mq)^x} = \frac{\phi(q)}{q^x} \sum_{m=1}^{\infty} \frac{1}{m^x}.$$

where the second equality follows from the definition of $\chi_0(n)$, and $\gcd(b, q) = 1$ if and only if $\gcd(mq+b, q) = 1$ for any $m \in \mathbb{N}_0$.

Thus, as $x \rightarrow 1^+$, we have

$$L(x; \chi_0) \geq \frac{\phi(q)}{q^x} \sum_{m=1}^{\infty} \frac{1}{m^x} \rightarrow \infty.$$

 **Exercise 1.6** The Principal Dirichlet Character χ_0 modulo q is defined as

$$\chi_0(n) = \begin{cases} 1 & \text{if } \gcd(n, q) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Show that

$$\chi_0(mn) = \chi_0(m)\chi_0(n) \quad \text{for all } m, n \in \mathbb{N}.$$

and

$$\chi_0(b+nq) = \chi_0(b) \quad \text{for all } b, n \in \mathbb{N}.$$

Solution First, we show that $\chi_0(mn) = \chi_0(m)\chi_0(n)$ for all $m, n \in \mathbb{N}$. We consider two cases:

(C1) If $\gcd(m, q) = 1$ and $\gcd(n, q) = 1$, then $\gcd(mn, q) = 1$. Therefore,

$$\chi_0(mn) = 1 = 1 \cdot 1 = \chi_0(m)\chi_0(n).$$

(C2) If either $\gcd(m, q) > 1$ or $\gcd(n, q) > 1$, then $\gcd(mn, q) > 1$. Therefore,

$$\chi_0(mn) = 0 = 0 \cdot \chi_0(n) = \chi_0(m)\chi_0(n) \quad \text{or} \quad \chi_0(mn) = 0 = \chi_0(m) \cdot 0 = \chi_0(m)\chi_0(n).$$

Thus, in both cases, we have $\chi_0(mn) = \chi_0(m)\chi_0(n)$.

Next, we show that $\chi_0(b+nq) = \chi_0(b)$ for all $b, n \in \mathbb{N}$. We consider two cases:

(C1) If $\gcd(b, q) = 1$, then $\gcd(b+nq, q) = 1$. Therefore,

$$\chi_0(b+nq) = 1 = \chi_0(b).$$

(C2) If $\gcd(b, q) > 1$, then $\gcd(b+nq, q) > 1$. Therefore,

$$\chi_0(b+nq) = 0 = \chi_0(b).$$

Thus, in both cases, we have $\chi_0(b+nq) = \chi_0(b)$.

Definition 1.8 (Dirichlet L-Function (Outline))

Given a Dirichlet character χ modulo q , the Dirichlet L-function $L(s, \chi)$ is defined for complex numbers s with $\Re(s) > 1$ by the series

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Similar to the Riemann zeta function, the Dirichlet L-function can be analytically continued to the entire complex plane except for a simple pole at $s = 1$ when χ is the principal character. 

1.4 Chebyshev's bounds

Lemma 1.4 (Legendre's Formula)

For any integer n and prime p , consider the prime factorization of $m!$:

$$m! = p^\alpha \cdot (\text{other prime factors not equal to } p).$$

Then we have

$$\alpha = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor.$$

and denote it as $\alpha(p) = \alpha$ of the prime p in the prime factorization of $m!$. 



Note The series $\sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor$ is actually finite since $\left\lfloor \frac{m}{p^k} \right\rfloor = 0$ for all $k > \log_p m$.

Proof Notice that the prime factors of $m!$ are exactly the all prime factors from 1 to m . Since

$$\left\lfloor \frac{m}{p^k} \right\rfloor = \#\{np^k : n \in \mathbb{N}, np^k \leq m\}$$

counts the number of multiples of p^k that are less than or equal to m , each multiple contributes at least k to the exponent α of p in the prime factorization of $m!$. Therefore, summing over all k gives the total exponent of p in $m!$:

$$\alpha = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor.$$



Exercise 1.7 Show that $2^n \leq \binom{2n}{n} < 4^n$ for all $n \in \mathbb{N}$.

Solution We can prove this inequality using the definition of the binomial coefficient:

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}.$$

We want to show that

$$2^n \leq \frac{(2n)!}{(n!)^2}.$$

Multiplying both sides by $(n!)^2$, we need to prove that

$$2^n (n!)^2 \leq (2n)!.$$

We can rewrite $(2n)!$ as the product of the first $2n$ integers:

$$(2n)! = 1 \cdot 2 \cdot 3 \cdots (2n).$$

Now, we can group the terms in pairs:

$$(2n)! = (1 \cdot 2) \cdot (3 \cdot 4) \cdots ((2n-1) \cdot (2n)).$$

Each pair $(2k-1) \cdot (2k)$ for $k = 1, 2, \dots, n$ is greater than or equal to $2k$, since

$$(2k-1) \cdot (2k) = 4k^2 - 2k = 2k(2k-1) \geq 2k \quad \text{for } k \geq 1.$$

Therefore,

$$(2n)! \geq (2 \cdot 1) \cdot (2 \cdot 2) \cdots (2 \cdot n) = 2^n (1 \cdot 2 \cdots n) = 2^n n!.$$

This shows that

$$2^n(n!)^2 \leq (2n)!.$$

For the upper bound, by using the binomial theorem, we have

$$4^n = (1+1)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} \geq \binom{2n}{n}.$$

Thus, we have shown that

$$2^n \leq \binom{2n}{n} < 4^n.$$

Lemma 1.5 (Chebyshev's Product Bounds lemma)

For n sufficiently large, we have

$$\left(\frac{4}{3}\right)^n \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} \quad \text{and} \quad \prod_{\substack{p \in \mathbb{P} \\ n < p \leq 2n}} p \leq 4^n.$$



Proof By Lemma 1.4, and notice that the prime factors of $m!$ are exactly the all prime factors from 1 to m , we have

$$n! = \prod_{\substack{p \in \mathbb{P} \\ p \leq n}} p^{\alpha(p)} \quad \text{where } \alpha(p) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Thus, by the definition of binomial coefficient, we have

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \frac{\prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\sum_{k=1}^{\infty} \left\lfloor \frac{2n}{p^k} \right\rfloor}}{\left(\prod_{\substack{p \in \mathbb{P} \\ p \leq n}} p^{\sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor} \right)^2} = \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\beta(p)}$$

where

$$\beta(p) = \sum_{k=1}^{\infty} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right).$$

By the **Note**, $\left\lfloor \frac{2n}{p^k} \right\rfloor = 0$ if $p^k > 2n$ or equivalently $k > \log_p(2n)$, and since $0 \leq \lfloor 2x \rfloor - 2\lfloor x \rfloor \leq 1$ for all $x \geq 0$, we have

$$\beta(p) \leq \sum_{k=1}^{\lfloor \log_p(2n) \rfloor} 1 = \lfloor \log_p(2n) \rfloor \leq \log_p(2n) \quad \text{and thus} \quad p^{\beta(p)} = p^{\log_p(2n)} \leq 2n.$$

Furthermore, when $p > \sqrt{2n}$, $\left\lfloor \frac{2n}{p^k} \right\rfloor = 0$ and $\left\lfloor \frac{n}{p^k} \right\rfloor = 0$ for all $k \geq 2$, so

$$\beta(p) = \left\lfloor \frac{2n}{p} \right\rfloor - 2 \left\lfloor \frac{n}{p} \right\rfloor \leq 1 \quad \text{for } p > \sqrt{2n}.$$

Hence,

$$\begin{aligned} \binom{2n}{n} &= \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\beta(p)} = \prod_{\substack{p \in \mathbb{P} \\ p \leq \sqrt{2n}}} p^{\beta(p)} \cdot \prod_{\substack{p \in \mathbb{P} \\ \sqrt{2n} < p \leq 2n}} p^{\beta(p)} \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq \sqrt{2n}}} 2n \cdot \prod_{\substack{p \in \mathbb{P} \\ \sqrt{2n} < p \leq 2n}} p \\ &\leq (2n)^{\sqrt{2n}} \cdot \prod_{\substack{p \in \mathbb{P} \\ \sqrt{2n} < p \leq 2n}} p \\ &\leq \left(\frac{3}{2}\right)^n \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p \end{aligned}$$

where the last inequality holds for n sufficiently large since

$$\lim_{n \rightarrow \infty} \frac{(2n)^{\sqrt{2n}}}{(3/2)^n} = \lim_{n \rightarrow \infty} \frac{e^{\sqrt{2n} \log(2n)}}{e^{n \log(3/2)}} = \lim_{n \rightarrow \infty} e^{n \left(\frac{\log(2n)}{\sqrt{2n}} - \log(3/2) \right)} = 0 \quad \text{since} \quad \lim_{n \rightarrow \infty} \frac{\log(2n)}{\sqrt{2n}} = \lim_{n \rightarrow \infty} \frac{1/n}{1/\sqrt{2n}} = 0.$$

where the last equality follows from L'Hôpital's rule if we treat n as a real variable and then replace it back to integer variable. Therefore, by Exercise 1.7, we have

$$2^n \leq \binom{2n}{n} \leq \left(\frac{3}{2}\right)^n \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p \implies \left(\frac{4}{3}\right)^n \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p.$$

For the second inequality, notice that

$$\beta(p) = \left\lfloor \frac{2n}{p} \right\rfloor - 2 \left\lfloor \frac{n}{p} \right\rfloor = 1 - 0 = 1 \quad \text{for } n < p \leq 2n,$$

Combining with Exercise 1.7 again, we have

$$4^n \geq \binom{2n}{n} = \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\beta(p)} \geq \prod_{\substack{p \in \mathbb{P} \\ n < p \leq 2n}} p.$$

Theorem 1.6 (Chebyshev's Bounds on $\pi(N)$)

There exist positive constants $c_1, c_2 > 0$ such that

$$c_1 \frac{N}{\log N} \leq \pi(N) \leq c_2 \frac{N}{\log N} \quad \text{for } N \text{ sufficiently large.}$$



Proof Chebyshev used the bounds in Lemma 1.5 to prove the theorem. By using the first inequality in Lemma 1.5, we have

$$\left(\frac{4}{3}\right)^n \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} 2n = (2n)^{\pi(2n)}.$$

Taking logarithm on both sides, we have

$$n \log \left(\frac{4}{3}\right) \leq \pi(2n) \log(2n) \implies \frac{\log(4/3)}{2} \cdot \frac{2n}{\log(2n)} \leq \pi(2n).$$

Thus, for $N = 2n$ sufficiently large, it gives the lower bound. And for $N = 2n + 1$ sufficiently large, we have

$$\pi(N) = \pi(2n + 1) \geq \pi(2n) \geq \frac{\log(4/3)}{2} \cdot \frac{2n}{\log(2n)} \geq \frac{\log(4/3)}{4} \cdot \frac{2n + 1}{\log(2n + 1)}$$

where the last inequality holds as $\frac{x}{\log x}$ is increasing for $x \geq e$ ($[\frac{x}{\log x}]' = \frac{\log x - 1}{(\log x)^2} > 0$ for $x > e$). Hence, we have the lower bound for all sufficiently large N with $c_1 = \frac{\log(4/3)}{4}$.

For the upper bound, by extend the second inequality in Lemma 1.5 (Let $n = \lfloor \frac{1}{2}x \rfloor, x > 0$), we have

$$\prod_{\substack{p \in \mathbb{P} \\ \lfloor x/2 \rfloor < p \leq 2\lfloor x/2 \rfloor}} p \leq 4^{\lfloor x/2 \rfloor} \leq 4^{x/2} = 2^x$$

and since there are at most one prime between $2\lfloor x/2 \rfloor$ and x . For x sufficiently large, we have

$$\left(\frac{x}{2}\right)^{\pi(x) - \pi(x/2)} \leq \left(\frac{x}{2}\right)^{\pi(x) - \pi(\lfloor x/2 \rfloor)} \leq \prod_{\substack{p \in \mathbb{P} \\ \lfloor x/2 \rfloor < p \leq x}} \frac{x}{2} \leq \prod_{\substack{p \in \mathbb{P} \\ \lfloor x/2 \rfloor < p \leq x}} p \leq x \cdot 2^x \leq 3^x.$$

Taking logarithm on both sides, we have

$$(\pi(x) - \pi(x/2)) \log \left(\frac{x}{2}\right) \leq x \log 3 \implies \pi(x) - \pi(x/2) \leq \frac{x \log 3}{\log(x/2)}.$$

Replacing x by $\frac{x}{2^{j-1}}$ for $j = 1, 2, \dots, m$

$$\pi\left(\frac{x}{2^{j-1}}\right) - \pi\left(\frac{x}{2^j}\right) \leq \frac{\frac{x}{2^{j-1}} \log 3}{\log\left(\frac{x}{2^{j-1}}\right)} \leq \log(9) \cdot \frac{x 2^{-j}}{\log(x 2^{-j})}.$$

Summing over $j = 1, 2, \dots, m$ gives

$$\pi(x) \leq \pi\left(\frac{x}{2^m}\right) + \log(9) \sum_{j=1}^m \frac{x 2^{-j}}{\log(x 2^{-j})}.$$

As for $j \leq m$, we have $\log(x 2^{-j}) \geq \log(x 2^{-m}) \geq \log(x)/2$, $\pi(\frac{x}{2^m}) \leq \frac{x}{2^m} \leq 2\sqrt{x}$ and $\sqrt{x} \leq \frac{x}{\log(x)}$ for x sufficiently

large, it gives

$$\pi(x) \leq 2\sqrt{x} + \log(9) \sum_{j=1}^m \frac{x2^{-j}}{\log(x)/2} = 2\sqrt{x} + \frac{2x \log(9)}{\log(x)} \sum_{j=1}^m 2^{-j} \leq 2\sqrt{x} + \frac{2x \log(9)}{\log(x)} \leq (1 + 2\log(9)) \frac{x}{\log(x)}$$

By setting $c_2 = 1 + 2\log(9)$, and $x = N$, we have the upper bound for all sufficiently large N .

Exercises

- For each $n \in \mathbb{N}$, show that the prime factors of $n(n+1)$ have more than the prime factors of n itself. And deduce that there are infinitely many primes by considering the sequence $\{x_n\}_{n=1}^{\infty}$ where $x_n = x_{n-1}(x_{n-1}+1)$. (Hint: show that $\gcd(n, n+1) = 1$.)

Solution By Euclid's algorithm, we have $\gcd(n, n+1) = \gcd(n+1-n, n) = \gcd(1, n) = 1$.

Thus, n and $n+1$ are coprime, which means they share no common prime factors. Therefore, the prime factors of $n(n+1)$ are the union of the prime factors of n and the prime factors of $n+1$, which means

$$\# \text{prime factors of } n(n+1) = \# \text{prime factors of } n + \# \text{prime factors of } (n+1).$$

and clearly, $\# \text{prime factors of } (n+1) \geq 1$ as it is either prime itself or has at least one prime factor. Hence, the prime factors of $n(n+1)$ have more than the prime factors of n itself.

For the sequence $\{x_n\}_{n=1}^{\infty}$ defined by $x_n = x_{n-1}(x_{n-1}+1)$ with $x_1 = 1$. By previous result, x_n has more (strictly) prime factors than x_{n-1} . Since $x_1 = 1$ has no prime factors, it follows that x_n has at least $n-1$ distinct prime factors. Therefore, as n increases, the number of distinct prime factors of x_n also increases without bound, which implies that there are infinitely many primes.

- Fix any positive integer N , let $P_k = 2, 3, 5, \dots, p_k$ denote the set of first k primes. Define

$$E_N^k = \{n \in \mathbb{N} : n \leq N \text{ and } n \text{ only divisible by any prime in } P_k\}.$$

Show that

- Every $n \in E_N^k$ can be expressed as

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} s^2$$

for some non-negative integers $e_1, e_2, \dots, e_k \in \{0, 1\}$ and some $s \in \mathbb{N}$.

- Then deduce that there are at most $2^k \sqrt{N}$ elements in E_N^k .
- Finally, for any fixed N , let $n \in \mathbb{N}$ such that $P_n = \{p_1, p_2, \dots, p_n\}$ will exact all the primes less than or equal to N . i.e.

$$P_n = \{p \in \mathbb{P} : p \leq N\} = \{2, 3, 5, \dots, p_n\}.$$

So that $\pi(N) = n$ and note that $E_N^n = \{1, 2, \dots, N\}$ (prove it!). From above results, we have $N = \#E_N^n \leq 2^n \sqrt{N}$ or equivalently

$$\pi(N) = n \geq \frac{\log N}{2 \log 2}.$$

By using this, we can show that $\pi(N) \rightarrow \infty$ as $N \rightarrow \infty$. And compare this bound with our previous bound for $\pi(N)$ in Proposition 1.1, which is

$$\pi(N) \geq \log \log N$$

Solution

- By the Fundamental Theorem of Arithmetic 1.1, every $n \in E_N^k$ can be expressed as

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

for some non-negative integers $a_1, a_2, \dots, a_k$. For each a_i , by using Euclid's division algorithm, we can write

$$a_i = 2d_i + r_i \quad \text{where } d_i \in \mathbb{N}_0 \text{ and } r_i \in \{0, 1\}.$$

Thus,

$$n = p_1^{2d_1+r_1} p_2^{2d_2+r_2} \cdots p_k^{2d_k+r_k} = (p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k})^2 \cdot p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}.$$

and we get the desired form by letting $s = p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k}$.

For the uniqueness of this representation, suppose there are two representations of n :

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} s^2 = p_1^{f_1} p_2^{f_2} \cdots p_k^{f_k} t^2$$

where $e_i, f_i \in \{0, 1\}$ and $s, t \in \mathbb{N}$. Rearranging gives

$$p_1^{e_1-f_1} p_2^{e_2-f_2} \cdots p_k^{e_k-f_k} = \left(\frac{t}{s}\right)^2.$$

The left side is an integer, so $\frac{t}{s}$ must also be an integer, say m . Thus,

$$p_1^{e_1-f_1} p_2^{e_2-f_2} \cdots p_k^{e_k-f_k} = m^2.$$

Since the right side is a perfect square, all exponents on the left side must be even. Given that $e_i, f_i \in \{0, 1\}$, the only way for $e_i - f_i$ to be even is if $e_i = f_i$ for all i . This implies that the exponents are unique.

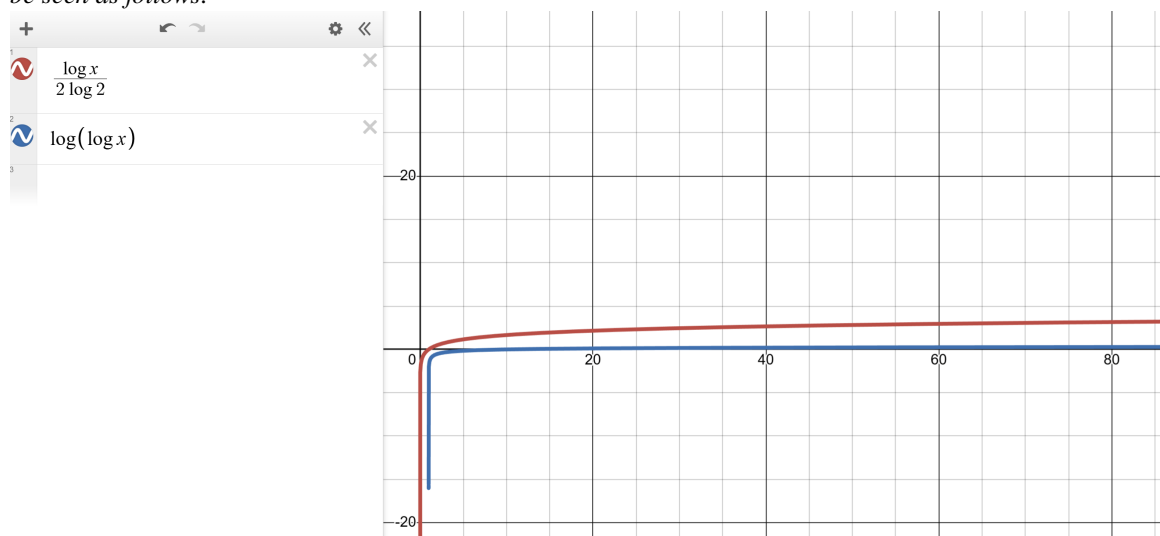
- (b) By the above result, for each $n \in E_N^k$, there are 2^k choices for the exponents $e_1, e_2, \dots, e_k$ and since $n \leq N$, we have

$$s^2 \leq n \leq N \implies s \leq \sqrt{N}.$$

Thus, there are at most $\sqrt{N}$ choices for s . Therefore, the total number of elements in E_N^k is at most $2^k \sqrt{N}$.

- (c) To show that $E_N^n = \{1, 2, \dots, N\}$, notice that any integer m such that $1 \leq m \leq N$ can only have prime factors less than or equal to N . Since P_n contains all primes less than or equal to N , it follows that every integer m in this range is included in E_N^n . Conversely, any number in E_N^n is by definition less than or equal to N . Thus, we have $E_N^n = \{1, 2, \dots, N\}$.

For the lower bound on $\pi(N)$, this bounds shows a quicker growth rate compared to the previous bound. Can be seen as follows:



3. This question show how previous exercises implies that $\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty$. Suppose by contradiction that $\sum_{p \in \mathbb{P}} \frac{1}{p}$ converges. Then the tails of the series must tend to zero, and so we can find a positive integer k such that

$$\sum_{i=k+1}^{\infty} \frac{1}{p_i} < \frac{1}{2}.$$

where we denote the i -th prime by p_i .

Now, fix a positive integer N and fix a prime p then we have

$$\#\{0 < n \leq N : p \mid n, n \in \mathbb{N}\} \leq \frac{N}{p}. \quad (\text{show it!})$$

Then consider E_N^k defined in the previous exercise, we have

$$N - \#E_N^k \leq \frac{N}{p_{k+1}} + \frac{N}{p_{k+2}} + \frac{N}{p_{k+3}} + \cdots < \frac{N}{2}. \quad (\text{show it!})$$

As $\#E_N^k \leq 2^k \sqrt{N}$ from previous exercise, we have

$$\frac{N}{2} < 2^k \sqrt{N} \implies N < 2^{2k+2}$$

which is impossible as N can be arbitrarily large. Hence, we conclude that $\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty$.

Solution

- (a) For a fixed prime p and positive integer N , the integers n such that $1 \leq n \leq N$ and $p \mid n$ are precisely the multiples of p : $p, 2p, 3p, \dots, mp$ where $mp \leq N$. Thus, the largest integer m satisfying this condition is $\lfloor \frac{N}{p} \rfloor$. Therefore we have the inequality.
- (b) For a integer $0 \leq n \leq N$, if $n \notin E_N^k$, then n must be divisible by at least one prime p_i where $i > k$. Thus, by the previous result, the number of integers n such that $1 \leq n \leq N$ and $n \notin E_N^k$ is at most $\frac{N}{p_{k+1}} + \frac{N}{p_{k+2}} + \frac{N}{p_{k+3}} + \dots$. By our assumption, this sum is less than $\frac{N}{2}$. Therefore, we have the inequality.
4. This is a topological way to show there are infinitely many primes. There are some basic concepts in topology we need to introduce first.

Suppose that a collection $\mathcal{B}$ of subsets of a set X (i.e. $\mathcal{B} \subseteq \mathcal{P}(X)$) satisfies the following two properties:

- (i) Every $x \in X$, there is at least one $B \in \mathcal{B}$ such that $x \in B$;
- (ii) For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there is a $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

then we call $\mathcal{B}$ a basis for a topology τ on X it generates. Here a open set of τ is either the empty set or it is a union of sets in $\mathcal{B}$ and a set $F \subset X$ is called closed if its complement $X \setminus F$ is open. We now consider the a topology on $X = \mathbb{Z}$.

- (a) Let $\mathcal{B} = \{S_{a,q} : a, q \in \mathbb{Z}, q > 0\}$ where $S_{a,q} = \{a + nq : n \in \mathbb{Z}\}$ denotes the arithmetic progression. Show that $\mathcal{B}$ is a basis for a topology τ on $\mathbb{Z}$ and any finite set $S \subset \mathbb{Z}$ cannot be open in this topology.
- (b) Show that $S_{a,q}$ is both open and closed for any $a, q \in \mathbb{Z}, q > 0$.
- (c) Finally, show that

$$\mathbb{Z} \setminus \{-1, 1\} = \bigcup_{p \in \mathbb{P}} S_{0,p}.$$

and thus if $\#\mathbb{P} < \infty$, then the right hand side is a closed set but the left hand side cannot be closed. Hence, we conclude that there are infinitely many primes.

Solution

- (a) For any $x \in \mathbb{Z}$, by letting $a = x$ and $q = 1$, then $x \in S_{a,q}$ and hence property (i) holds. For any $B_1 = S_{a_1,q_1}, B_2 = S_{a_2,q_2} \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, then $x = a_1 + n_1q_1 = a_2 + n_2q_2$ and by letting $q_3 = \text{lcm}(q_1, q_2)$ and $a_3 = x$. Then $x \in S_{a_3,q_3}$ and for any $y \in S_{a_3,q_3}$, we have

$$y = a_3 + n_3q_3 = x + n_3\text{lcm}(q_1, q_2) = x + n_3m_1q_1 = x + n'_3m_2q_2$$

for some integers m_1, m_2, n_3, n'_3 . Thus, $y \in B_1 \cap B_2$ and hence property (ii) holds. Therefore, $\mathcal{B}$ is a basis for a topology τ on $\mathbb{Z}$.

For any finite set $S \subset \mathbb{Z}$. Suppose by contradiction that S is open, then S is either the empty set (which is not the case here) or it is a union of sets in $\mathcal{B}$. Since each set in $\mathcal{B}$ is infinite, any union of sets in $\mathcal{B}$ must also be infinite. This contradicts the assumption that S is finite. Hence, any finite set $S \subset \mathbb{Z}$ cannot be open in this topology.

- (b) For any $S_{a,q} \in \mathcal{B}$, it is open as it is a union of itself. To show that it is closed, consider its complement

$$\mathbb{Z} \setminus S_{a,q} = \bigcup_{r=0}^{q-1} S_{r,q} \setminus S_{a,q} = \bigcup_{\substack{r=0 \\ r \neq a \pmod q}}^{q-1} S_{r,q}$$

which is a union of sets in $\mathcal{B}$ and hence is open. Therefore, $S_{a,q}$ is closed.

- (c) for each $n \in \mathbb{Z} \setminus \{-1, 1\}$, if $n = 0$, then $n \in S_{0,2}$; if $|n| > 1$, then by the Fundamental Theorem of Arithmetic 1.1, n has at least one prime factor p , so that $n \in S_{0,p}$. Thus $\mathbb{Z} \setminus \{-1, 1\} \subseteq \bigcup_{p \in \mathbb{P}} S_{0,p}$. Conversely, for any $n \in \bigcup_{p \in \mathbb{P}} S_{0,p}$, then there exists a prime p such that $n \in S_{0,p}$, which means $p \mid n$. Therefore, n cannot be ± 1 . Thus, $\bigcup_{p \in \mathbb{P}} S_{0,p} \subseteq \mathbb{Z} \setminus \{-1, 1\}$. Therefore, we get the desired equality.

Chapter 2 Infinite Products and Series

2.1 Infinite Products, Uniform Convergence Criteria and M-test

Definition 2.1 (Infinite Product)

A infinite product $\prod_{n=1}^{\infty} a_n$ converges if the sequence of partial products

$$P_N = \prod_{n=1}^N a_n$$

converges to a limit P as $N \rightarrow \infty$. In this case, we write

$$\prod_{n=1}^{\infty} a_n = P.$$



Theorem 2.1 (n-term Test, Divergence Test for Infinite Products)

If the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit, then $\lim_{n \rightarrow \infty} a_n = 1$.

And thus we usually write infinite products in the form

$$\prod_{n=1}^{\infty} (1 + b_n)$$

and then the n-term test becomes: if $\prod_{n=1}^{\infty} (1 + b_n)$ converges, then $\lim_{n \rightarrow \infty} b_n = 0$.



Proof Suppose that the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit P . Then the sequence of partial products

$$P_N = \prod_{n=1}^N a_n \rightarrow P \neq 0 \quad \text{as } N \rightarrow \infty.$$

Then, $a_n \neq 0$ for any $n \in \mathbb{N}$, and we have

$$a_n = \frac{P_n}{P_{n-1}} \rightarrow \frac{P}{P} = 1 \quad \text{as } n \rightarrow \infty.$$

Definition 2.2 (Bounded Complex-Value Function)

A complex-value function $G : X \rightarrow \mathbb{C}$ defined on a set X is bounded if

$$\|G\|_{\infty} := \sup_{x \in X} |G(x)| < \infty.$$

and the number $\|G\|_{\infty}$ is the least upper bound of $|G(x)|$ for all $x \in X$.



Definition 2.3

A sequence of complex-value functions $S_n : X \rightarrow \mathbb{C}$ converges uniformly to a complex-value function $S : X \rightarrow \mathbb{C}$ on a set X if

$$\lim_{n \rightarrow \infty} \|S_n - S\|_{\infty} = \lim_{n \rightarrow \infty} \sup_{x \in X} |S_n(x) - S(x)| = 0.$$



Exercise 2.1 If a sequence of complex-value functions $S_n : X \rightarrow \mathbb{C}$ converges uniformly to a complex-value function $S : X \rightarrow \mathbb{C}$ on a set X , then S is a bounded function on X .

Solution Since S_n converges uniformly to S on X , for any $\varepsilon > 0$, there exists an integer N such that for all $n \geq N$,

$$\|S_n - S\|_{\infty} = \sup_{x \in X} |S_n(x) - S(x)| < \varepsilon.$$

This implies that for all $x \in X$ and all $n \geq N$,

$$|S(x)| \leq |S_n(x)| + |S_n(x) - S(x)| < |S_n(x)| + \varepsilon.$$

Since S_N is a bounded function on X , there exists a constant M_N such that

$$|S_N(x)| \leq M_N \quad \text{for all } x \in X.$$

Therefore, for all $x \in X$,

$$|S(x)| < M_N + \varepsilon.$$

Thus,

$$\sup_{x \in X} |S(x)| \leq M_N + \varepsilon < \infty.$$

Hence, S is a bounded function on X .

 **Exercise 2.2** Show that if $F, G : X \rightarrow \mathbb{C}$ are bounded complex-value functions on a set X , then $\|F + G\|_\infty \leq \|F\|_\infty + \|G\|_\infty$ and $\|FG\|_\infty \leq \|F\|_\infty \|G\|_\infty$.

Solution For any $x \in X$, by the triangle inequality, we have

$$|F(x) + G(x)| \leq |F(x)| + |G(x)| \leq \|F\|_\infty + \|G\|_\infty.$$

Taking the supremum over all $x \in X$, we get

$$\|F + G\|_\infty = \sup_{x \in X} |F(x) + G(x)| \leq \|F\|_\infty + \|G\|_\infty.$$

Similarly, for the product, we have

$$|F(x)G(x)| = |F(x)| |G(x)| \leq \|F\|_\infty \|G\|_\infty.$$

Taking the supremum over all $x \in X$, we get

$$\|FG\|_\infty = \sup_{x \in X} |F(x)G(x)| \leq \|F\|_\infty \|G\|_\infty.$$

Definition 2.4 (Uniform Convergence of Series of Functions)

A series of complex-value bounded functions $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the sequence of partial sums

$$S_N(x) = \sum_{n=1}^N f_n(x)$$

converges uniformly on X . 

Definition 2.5 (Uniform Convergence of Products of Functions)

A product of complex-value bounded functions $\prod_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the sequence of partial products

$$P_N(x) = \prod_{n=1}^N f_n(x)$$

converges uniformly on X . 

Theorem 2.2 (Cauchy Criterion for Uniform Convergence of Series of Functions)

A series of complex-value bounded functions $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges.

$$S_N(x) = \sum_{n=1}^N f_n(x)$$

satisfies

$$\|S_N - S\|_\infty \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$



Proof For any $\varepsilon > 0$, since the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges, there exists an integer N such that for all

$m > n \geq N$,

$$\sum_{k=n+1}^m \|f_k\|_\infty < \varepsilon.$$

Then, for any $x \in X$ and all $m > n \geq N$, we have

$$|S_m(x) - S_n(x)| = \left| \sum_{k=n+1}^m f_k(x) \right| \leq \sum_{k=n+1}^m |f_k(x)| \leq \sum_{k=n+1}^m \|f_k\|_\infty < \varepsilon.$$

Taking the supremum over all $x \in X$, we get

$$\|S_m - S_n\|_\infty = \sup_{x \in X} |S_m(x) - S_n(x)| < \varepsilon.$$

This shows that the sequence of partial sums $\{S_N\}$ is uniformly Cauchy on X . Since the space of bounded functions on X is complete with respect to the supremum norm, it follows that $\{S_N\}$ converges uniformly to a bounded function $S : X \rightarrow \mathbb{C}$ on X . Hence, the series $\sum_{n=1}^\infty f_n(x)$ converges uniformly on X .

Theorem 2.3

A product of complex-value bounded functions $\prod_{n=1}^\infty (1 + f_n(x))$ converges uniformly on a set X if the series of norms $\sum_{n=1}^\infty \|f_n\|_\infty$ converges.

$$P_N(x) = \prod_{n=1}^N (1 + f_n(x))$$

satisfies

$$\|P_N - P\|_\infty \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Proof Let

$$P_N(x) = \prod_{k=1}^N (1 + f_k(x)), \quad x \in X.$$

Fix $\varepsilon > 0$. Since $\sum_{n=1}^\infty \|f_n\|_\infty$ converges, define

$$A := \sum_{n=1}^\infty \|f_n\|_\infty < \infty, \quad M := e^A.$$

For any $n \in \mathbb{N}$ and any $x \in X$, we have

$$|P_n(x)| = \prod_{k=1}^n |1 + f_k(x)| \leq \prod_{k=1}^n (1 + |f_k(x)|) \leq \prod_{k=1}^n (1 + \|f_k\|_\infty) \leq \exp\left(\sum_{k=1}^n \|f_k\|_\infty\right) \leq M.$$

Now choose $\delta > 0$ such that

$$e^\delta - 1 < \frac{\varepsilon}{M}.$$

Since $\sum_{n=1}^\infty \|f_n\|_\infty$ converges, there exists $N \in \mathbb{N}$ such that for all $m > n \geq N$,

$$\sum_{k=n+1}^m \|f_k\|_\infty < \delta.$$

For any $x \in X$ and $m > n \geq N$, we write

$$|P_m(x) - P_n(x)| = \left| \prod_{k=1}^n (1 + f_k(x)) \left(\prod_{k=n+1}^m (1 + f_k(x)) - 1 \right) \right|.$$

Using the inequality $|1 + z| \leq e^{|z|}$ for $z \in \mathbb{C}$, we obtain

$$\left| \prod_{k=n+1}^m (1 + f_k(x)) \right| \leq \exp\left(\sum_{k=n+1}^m |f_k(x)|\right),$$

and hence

$$\left| \prod_{k=n+1}^m (1 + f_k(x)) - 1 \right| \leq e^{\sum_{k=n+1}^m |f_k(x)|} - 1 < e^\delta - 1.$$

Combining the estimates, we obtain

$$|P_m(x) - P_n(x)| \leq M(e^\delta - 1) < \varepsilon.$$

Taking the supremum over all $x \in X$, we get

$$\|P_m - P_n\|_\infty = \sup_{x \in X} |P_m(x) - P_n(x)| < \varepsilon.$$

Thus $\{P_N\}$ is a uniformly Cauchy sequence on X . Since the space of bounded functions on X is complete with respect to the supremum norm, there exists a bounded function $P : X \rightarrow \mathbb{C}$ such that

$$\|P_N - P\|_\infty \rightarrow 0 \quad (N \rightarrow \infty).$$

Therefore, the infinite product $\prod_{n=1}^\infty (1 + f_n(x))$ converges uniformly on X .

Theorem 2.4 (Uniform Convergence and Zeros of Infinite Products)

Let $u_n : X \rightarrow \mathbb{C}$ be a sequence of bounded complex-value functions on a set X such that the series

$$\sum_{n=1}^\infty |u_n| \text{ converges uniformly on } X.$$

Then the infinite product

$$f(x) = \prod_{n=1}^\infty (1 + u_n(x))$$

converges uniformly on X .

Furthermore,

$$f(x_0) = 0 \text{ for some } x_0 \in X \iff u_n(x_0) = -1 \text{ for some } n \in \mathbb{N}.$$

i.e. f has a zero at x_0 if and only if one of the factors in the product is zero at x_0 .



Note We can use this theorem even in the case where u_n is constant function. And in this case, it states that

$$\sum_{n=1}^\infty |u_n| \text{ converges and } u_n \neq -1 \text{ for all } n \in \mathbb{N} \implies \prod_{n=1}^\infty (1 + u_n) \text{ converges to a nonzero limit.}$$

Exercise 2.3 Show that $e^x \geq x + 1$ for all $x \geq 0$

Solution By applying Mean Value Theorem to the function $f(x) = e^x$ on the interval $[0, x]$ for any $x \geq 0$, there exists some $c \in (0, x)$ such that

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \implies e^c = \frac{e^x - 1}{x}.$$

Since $e^c \geq 1$ for all $c \geq 0$, we have

$$\frac{e^x - 1}{x} \geq 1 \implies e^x - 1 \geq x \implies e^x \geq x + 1.$$

Exercise 2.4 Show that for any sequence of complex numbers $\{a_n\}_{n=1}^\infty$, we have

$$\left| \prod_{n=1}^N (1 + a_n) - 1 \right| \leq \exp \left(\sum_{n=1}^N |a_n| \right) - 1.$$

Solution We prove this by induction on N . For $N = 1$, we have

$$|(1 + a_1) - 1| = |a_1| \leq e^{|a_1|} - 1, \text{ by Exercise 2.3.}$$

Now, assume the inequality holds for some $N \geq 1$. Then for $N + 1$, we have

$$\begin{aligned}
 \left| \prod_{n=1}^{N+1} (1 + a_n) - 1 \right| &= \left| (1 + a_{N+1}) \prod_{n=1}^N (1 + a_n) - 1 \right| = \left| (1 + a_{N+1}) \left(\prod_{n=1}^N (1 + a_n) - 1 \right) + a_{N+1} \right| \\
 &\leq |1 + a_{N+1}| \left| \prod_{n=1}^N (1 + a_n) - 1 \right| + |a_{N+1}| \\
 &\leq (1 + |a_{N+1}|) \left(e^{\sum_{n=1}^N |a_n|} - 1 \right) + |a_{N+1}|, \text{ by the induction hypothesis} \\
 &= e^{\sum_{n=1}^N |a_n|} - 1 + |a_{N+1}| e^{\sum_{n=1}^N |a_n|} - |a_{N+1}| + |a_{N+1}| \\
 &= (1 + |a_{N+1}|) e^{\sum_{n=1}^N |a_n|} - 1 \\
 &\leq e^{|a_{N+1}|} e^{\sum_{n=1}^N |a_n|} - 1, \text{ by Exercise 2.3} \\
 &= e^{\sum_{n=1}^{N+1} |a_n|} - 1.
 \end{aligned}$$

Thus, by induction, the inequality holds for all $N \in \mathbb{N}$.

Proof By our assumption, as $\sum_{n=1}^{\infty} |u_n|$ converges uniformly on X , the partial sums $S_N(x) = \sum_{n=1}^N |u_n(x)|$ converge uniformly to $S = \sum_{n=1}^{\infty} |u_n(x)|$ on X . Thus, for a given $\varepsilon > 0$, there is a N_0 (not depending on x) such that

$$\|S_N - S\|_{\infty} = \sup_{x \in X} \sum_{n=N+1}^{\infty} |u_n(x)| < \varepsilon, \quad \text{for all } N \geq N_0.$$

and exercise 2.1 implies that S is bounded on X . Hence, $S(x) = \sum_{n=1}^{\infty} |u_n(x)| \leq M$ for some $M > 0$ and all $x \in X$.

Let $P_N(x) = \prod_{n=1}^N (1 + u_n(x))$ denote the N -th partial product of $f(x)$. Then by exercise 2.4, we have

$$|P_N(x) - 1| \leq \exp \left(\sum_{n=1}^N |u_n(x)| \right) - 1 \leq e^M - 1 \quad \text{since } \sum_{n=1}^N |u_n(x)| \leq S(x) \leq M.$$

and thus $|P_N(x)| \leq e^M - 1 + 1 = e^M$ for all $x \in X$ and all $N \in \mathbb{N}$. Therefore, for any $M > N \geq N_0$ and any $x \in X$, using exercise 2.4 again, we have

$$|P_M(x) - P_N(x)| = |P_N(x)| \left| \prod_{n=N+1}^M (1 + u_n(x)) - 1 \right| \leq e^M \left(\exp \left(\sum_{n=N+1}^M |u_n(x)| \right) - 1 \right) < e^M (e^{\varepsilon} - 1) \quad (2.1)$$

for all $x \in X$. Taking the supremum over all $x \in X$, we get

$$\|P_M - P_N\|_{\infty} = \sup_{x \in X} |P_M(x) - P_N(x)| < e^M (e^{\varepsilon} - 1) < \varepsilon'$$

when we choose $\varepsilon = \log \left(\frac{\varepsilon'}{e^M} + 1 \right) > 0$. Thus, $\{P_N\}$ is a uniformly Cauchy sequence on X , and hence $f(x)$ converges uniformly on X .

Now, do the similar argument as in equation (2.1) and choose $M \rightarrow \infty$, $N = N_0$, we have

$$|f(x) - P_{N_0}(x)| \leq |P_{N_0}|(e^{\varepsilon} - 1) \quad \text{for all } x \in X.$$

Setting $\varepsilon < \frac{1}{2}$, we have

$$|f(x) - P_{N_0}(x)| < |P_{N_0}(x)| |\sqrt{e} - 1|$$

thus,

$$|f(x)| = |P_{N_0}(x) - (f(x) - P_{N_0}(x))| \geq |P_{N_0}(x)| - |P_{N_0}(x)| |\sqrt{e} - 1| = |P_{N_0}(x)| (2 - \sqrt{e}) \geq \frac{1}{3} |P_{N_0}(x)|$$

Therefore, if $f(x_0) = 0$ for some $x_0 \in X$, then $P_{N_0}(x_0) = 0$ and hence $u_n(x_0) = -1$ for some $1 \leq n \leq N_0$. The converse is trivial.

Theorem 2.5 (Weierstrass M-test for Infinite Series and Products)

If a sequence of complex function $u_n : X \rightarrow \mathbb{C}$ satisfies $|u_n(x)| \leq M_n$ and $\sum_{n=1}^{\infty} M_n$ converges, then the infinite series

$$\sum_{n=1}^{\infty} |u_n(x)| \text{ converges uniformly on } X.$$

hence the series

$$\sum_{n=1}^{\infty} u_n(x) \text{ converges uniformly on } X.$$

And by the previous theorem 2.4, the infinite product

$$f(x) = \prod_{n=1}^{\infty} (1 + u_n(x))$$

converges uniformly on X .



Proof Direct from the M -test in Complex Analysis and theorem 2.4.

Proposition 2.1 (Properties of Uniform Convergence)

1. uniform limit of continuous functions is continuous.
2. uniform limit of bounded functions is bounded. (See exercise 2.1.)
3. uniform limit of analytic functions is analytic.



Proof omitted.

Corollary 2.1 (Analyticity of Riemann Zeta Function and Euler Product)

Recall the Riemann zeta function and the Euler product formula

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad \text{and} \quad \mathcal{P}(z) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}} = \prod_{p \in \mathbb{P}} (1 + u_p(z)) \quad \text{for } \Re(z) > 1,$$

where $u_p(z) = \frac{1}{1-p^z} - 1 = \frac{p^{-z}}{1-p^{-z}}$.

Then both $\zeta(z)$ and $\mathcal{P}(z)$ are analytic on the half plane $\{z \in \mathbb{C} : \Re(z) > 1\}$. Further more, if

$$\zeta(z) = \mathcal{P}(z) \quad \text{for } \Re(z) > 1,$$

we can show that

$$|\zeta(z)| \neq 0 \quad \text{on } \{z \in \mathbb{C} : \Re(z) > 1\}$$

(i.e., $\zeta(z)$ has no zero on this half plane)



Note For a^z , where $a > 0$ is a positive real number, if not specified, we always consider it as $e^{z \log a}$, where $\log a$ is using the real logarithm to avoid multi-valued issue. And thus a^z is analytic on the whole complex plane $\mathbb{C}$.

Proof Notice that $\frac{1}{n^z}$ is an analytic function on $\mathbb{C}$ and $u_p(z)$ is analytic on $\{z \in \mathbb{C} : \Re(z) > 1\}$ for each prime p . Also,

$$|1 - p^{-z}| \geq 1 - |p^{-z}| = 1 - p^{-\Re(z)} \geq 1 - p^{-1} \geq \frac{1}{2} \quad \text{for } \Re(z) > 1.$$

Thus, for $\Re(z) \geq 1$, on the open set $X_\varepsilon = \{z \in \mathbb{C} : \Re(z) \geq 1 + \varepsilon\}$ for any $\varepsilon > 0$, we have

$$|u_p(z)| = \left| \frac{p^{-z}}{1 - p^{-z}} \right| \leq 2|p^{-z}| = 2p^{-\Re(z)} \leq 2p^{-(1+\varepsilon)} \quad \text{and} \quad \left| \frac{1}{n^z} \right| = n^{-\Re(z)} \leq n^{-(1+\varepsilon)}.$$

Since

$$\sum_{p \in \mathbb{P}} \frac{1}{p^{1+\varepsilon}} \leq \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} < \infty$$

the Weierstrass M-test 2.5 implies that both $\zeta(z)$ and $\mathcal{P}(z)$ converge uniformly on X_ε for any $\varepsilon > 0$. Therefore, by proposition 2.1, both $\zeta(z)$ and $\mathcal{P}(z)$ are analytic on $\{z \in \mathbb{C} : \Re(z) > 1\}$.

Further more, by theorem 2.4, since $u_p(z) \neq -1$ for all primes p and all z with $\Re(z) > 1$, we have $|\mathcal{P}(z)| \neq 0$ for all z with $\Re(z) > 1$. As $\zeta(z) = \mathcal{P}(z)$ for $\Re(z) > 1$, we have $|\zeta(z)| = |\mathcal{P}(z)| \neq 0$ for all z with $\Re(z) > 1$.

Note The same analysis can be applied to Dirichlet L -functions and their Euler product representations

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} \quad \text{and} \quad \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-z}}$$

but will discuss later once we have introduced Dirichlet characters.

2.2 Recap of Complex Analysis

Theorem 2.6 (Morera's Theorem)

Suppose $f : X \rightarrow \mathbb{C}$ is continuous function on an open set $X \subseteq \mathbb{C}$. If

$$\int_{\Gamma} f(z) dz = 0 \quad \text{for all loop } \Gamma \text{ in } X, \quad \text{then } f \text{ is analytic on } X.$$



Proof Omitted.

Theorem 2.7 (Cauchy Integral Theorem)

Suppose $f : X \rightarrow \mathbb{C}$ is an analytic function inside and on a simple closed contour Γ in an open set $X \subseteq \mathbb{C}$. Then

$$\int_{\Gamma} f(z) dz = 0$$



Proof Omitted.

Exercise 2.5 Use Morera's Theorem and Cauchy Integral Theorem to show the 3rd property in the above proposition 2.1.

Solution Suppose $\{f_n\}$ is a sequence of analytic functions on an open set $X \subseteq \mathbb{C}$ that converges uniformly to a function f on X . We want to show that f is analytic on X .

For any simple closed contour Γ in X , as f_n is analytic on X , by Cauchy Integral Theorem, we have

$$\int_{\Gamma} f_n(z) dz = 0 \quad \text{for all } n \in \mathbb{N}.$$

Since f_n converges uniformly to f on X , by the properties of uniform convergence, we can interchange the limit and the integral:

$$\int_{\Gamma} f(z) dz = \lim_{n \rightarrow \infty} \int_{\Gamma} f_n(z) dz = \lim_{n \rightarrow \infty} 0 = 0.$$

Thus, by Morera's Theorem, since f is continuous (as uniform limit of continuous functions is continuous) and the integral of f over any simple closed contour in X is zero, we conclude that f is analytic on X .

2.3 Euler Products

Definition 2.6

For a function $f : \mathbb{N} \rightarrow \mathbb{C}$ which is not identically zero, we say:

- (a) f is multiplicative, if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$.
- (b) f is completely multiplicative, if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$.



Exercise 2.6 Show that the Möbius function $\mu(n)$ which is defined as

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 \cdots p_k \text{ is a product of } k \text{ distinct primes,} \\ 0 & \text{otherwise,} \end{cases}$$

is a multiplicative function.

Solution Let $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. We want to show that $\mu(mn) = \mu(m)\mu(n)$.

Case 1: If either m or n has a squared prime factor, then mn also has a squared prime factor. Thus, $\mu(m) = 0$ or $\mu(n) = 0$, and hence $\mu(mn) = 0 = \mu(m)\mu(n)$.

Case 2: If both m and n are products of distinct primes, say $m = p_1 p_2 \cdots p_k$ and $n = q_1 q_2 \cdots q_l$, where p_i and q_j are distinct primes. Since $\gcd(m, n) = 1$, the sets of primes $\{p_1, p_2, \dots, p_k\}$ and $\{q_1, q_2, \dots, q_l\}$ are disjoint. Therefore,

$mn = p_1 p_2 \cdots p_k q_1 q_2 \cdots q_l$ is a product of $k + l$ distinct primes. Thus,

$$\mu(mn) = (-1)^{k+l} = (-1)^k (-1)^l = \mu(m)\mu(n).$$

Case 3: If either m or n is equal to 1, then without loss of generality, let $m = 1$. Then $\mu(1) = 1$, and we have

$$\mu(mn) = \mu(n) = \mu(1)\mu(n).$$

In all cases, we have shown that $\mu(mn) = \mu(m)\mu(n)$ for all $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. Therefore, the Möbius function $\mu(n)$ is multiplicative.

Proposition 2.2

If $f : \mathbb{N} \rightarrow \mathbb{C}$ is a (completely) multiplicative function, and $f(n_0) \neq 0$ for some $n_0 \in \mathbb{N}$, then $f(1) = 1$.



 **Note** If f is identically zero, then $f(mn) = f(m)f(n)$ holds trivially for all $m, n \in \mathbb{N}$. Thus, if not specified, we always assume that a (completely) multiplicative function is not identically zero. And then by the above proposition holds always for any (completely) multiplicative function.

 **Exercise 2.7** Prove the above proposition.

Solution Since f is (completely) multiplicative, we have

$$f(n_0) = f(1 \cdot n_0) = f(1)f(n_0) \quad \text{since } \gcd(1, n_0) = 1.$$

Since $f(n_0) \neq 0$, we can divide both sides by $f(n_0)$ to get

$$f(1) = 1.$$

Lemma 2.1 (“Euler Product”)

Suppose $f : \mathbb{N} \rightarrow \mathbb{C}$ is a multiplicative function (not identically zero) and the series

$$\sum_{n=1}^{\infty} |f(n)| < \infty.$$

Then the “Euler Product” holds:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p \in \mathbb{P}} (1 + f(p) + f(p^2) + \cdots) = \prod_{p \in \mathbb{P}} \sum_{k=0}^{\infty} f(p^k).$$



 **Note** The last equality holds since f is multiplicative function (not identically zero) implies that $f(1) = 1$ by proposition 2.2.

Proof Let $u_p = \sum_{k \geq 1} f(p^k)$ for each prime p and set

$$P_N = \prod_{\substack{p \in \mathbb{P} \\ p \leq N}} (1 + u_p) = (1 + u_{p_1})(1 + u_{p_2}) \cdots (1 + u_{p_r}),$$

where $p_1, p_2, \dots, p_r$ are all the primes less than or equal to N and suppose $p_1 < p_2 < \cdots < p_r$. We need to show u_p is convergent for each prime p to ensure P_N is well-defined (i.e. The limit of P_N exists) for each N . As $|u_p| \leq \sum_{k=1}^{\infty} |f(p^k)| \leq \sum_{n=1}^{\infty} |f(n)| < \infty$, each u_p is convergent.

Notice that $f(1) = 1$ by proposition 2.2. Thus,

$$P_N = \lim_{K \rightarrow \infty} \left[\sum_{k_1=0}^K f(p_1^{k_1}) \sum_{k_2=0}^K f(p_2^{k_2}) \cdots \sum_{k_r=0}^K f(p_r^{k_r}) \right].$$

because if $\lim_{K \rightarrow \infty} a_K = A$ and $\lim_{K \rightarrow \infty} b_K = B$, then $\lim_{K \rightarrow \infty} a_K b_K = AB$. Rearranging the sums on the RHS, and using the multiplicative property of f , we have

$$\sum_{k_1=0}^K \sum_{k_2=0}^K \cdots \sum_{k_r=0}^K f(p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}) = \sum_{n \in F_{N,K}} f(n) \quad \text{since } \gcd(p_i^{k_i}, p_j^{k_j}) = 1 \text{ for } i \neq j.$$

where

$$F_{N,K} = \{n \in \mathbb{N} : n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}, 0 \leq k_i \leq K \text{ for } 1 \leq i \leq r\}.$$

Recall u_p is convergent, P_N exists for each N . Thus

$$\lim_{K \rightarrow \infty} \sum_{n \in F_{N,K}} f(n) = P_N.$$

Since $F_{N,K} \subseteq \mathbb{N}$, we can choose a large $M_{N,K}$ such that $F_{N,K} \subseteq \{1, 2, \dots, M_{N,K}\}$ for each N, K . Then

$$\left| \sum_{n=1}^{M_{N,K}} f(n) - \sum_{n \in F_{N,K}} f(n) \right| = \left| \sum_{\substack{n \leq M_{N,K} \\ n \notin F_{N,K}}} f(n) \right| \leq \sum_{n \geq N} |f(n)| + \sum_{n \geq 2^K} |f(n)| \quad (2.2)$$

where the last inequality holds since any $n \notin F_{N,K}$, it is either has a prime factor greater than N (So that $n > N$) or has a prime factor less than or equal to N with exponent greater than K (So that $n \geq p_i^K \geq 2^K$ for some $1 \leq i \leq r$).

Fix a value of N , and let $K \rightarrow \infty$, notice that

$$\lim_{K \rightarrow \infty} \sum_{n=1}^{M_{N,K}} f(n) = \sum_{n=1}^{\infty} f(n) \quad \text{and} \quad \lim_{K \rightarrow \infty} \sum_{n \in F_{N,K}} f(n) = P_N.$$

Thus, taking $K \rightarrow \infty$ in the above inequality (2.2), we obtain

$$\left| \sum_{n=1}^{\infty} f(n) - P_N \right| \leq \sum_{n \geq N} |f(n)|.$$

Finally, letting $N \rightarrow \infty$. As $\sum_{n=1}^{\infty} |f(n)|$ converges, the tail $\sum_{n \geq N} |f(n)| \rightarrow 0$ as $N \rightarrow \infty$. Thus,

$$\sum_{n=1}^{\infty} f(n) = \lim_{N \rightarrow \infty} P_N = \prod_{p \in \mathbb{P}} (1 + f(p) + f(p^2) + \dots).$$

Theorem 2.8 (Euler Product Formula)

Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be a completely multiplicative function (not identically zero) and suppose the series

$$\sum_{n=1}^{\infty} |f(n)| < \infty.$$

Then the Euler Product holds:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p \in \mathbb{P}} (1 + f(p) + f(p^2) + \dots) = \prod_{p \in \mathbb{P}} \sum_{k=0}^{\infty} f(p)^k = \prod_{p \in \mathbb{P}} \frac{1}{1 - f(p)}.$$

Proof We only need to prove the last equality holds as the first two equalities are direct from lemma 2.1.

Recall the geometric series formula,

$$1 + z + z^2 + \dots = \frac{1}{1 - z} \quad \text{for } |z| < 1.$$

one may want to use this to justify the last equality. However, we need to ensure that $|f(p)| < 1$ for all primes p to apply the geometric series formula. But this is indeed true since if there exists a prime p_0 such that $|f(p_0)| \geq 1$, then as f is completely multiplicative, we have

$$|f(p_0^k)| = |f(p_0)|^k \geq 1 \quad \text{for all } k \in \mathbb{N},$$

thus the series $\sum_{n=1}^{\infty} |f(n)|$ diverges, as the subseries $\sum_{k=1}^{\infty} |f(p_0^k)|$ diverges. This contradicts our assumption that $\sum_{n=1}^{\infty} |f(n)| < \infty$. Therefore, we must have $|f(p)| < 1$ for all primes p , and hence we can apply the geometric series formula to get the last equality.

Corollary 2.2 (Classical Euler Product Formula)

The Riemann zeta function and the Euler product formula

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad \text{and} \quad \mathcal{P}(z) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}} \quad \text{for } \Re(z) > 1,$$

are equal, i.e.,

$$\zeta(z) = \mathcal{P}(z) \quad \text{for } \Re(z) > 1.$$



Proof By applying theorem 2.8 to the completely multiplicative function $f(n) = \frac{1}{n^z}$ for $\Re(z) > 1$, and noticing that $\sum_{n=1}^{\infty} |f(n)| = \sum_{n=1}^{\infty} \frac{1}{n^{\Re(z)}}$ converges for $\Re(z) > 1$, we have

$$\sum_{n=1}^{\infty} \frac{1}{n^z} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{1}{p^z}} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}}.$$

Remark We will also want to apply $f(n) = \chi(n)n^{-z}$ to get the Euler product formula for Dirichlet L -functions later, where χ is a Dirichlet character. And extend the Euler product formula to

$$L(z, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-z}} \quad \text{for } \Re(z) > 1.$$

2.4 Taking Logarithm

Definition 2.7 (Simple Connected)

An open set $X \subseteq \mathbb{C}$ is said to be simply connected if there is no hole in X , i.e., any loop in X can be continuously contracted to a point within X .



Theorem 2.9 (Existence of Logarithm Branch)

Let $X \subseteq \mathbb{C}$ be a simply connected open set, and $g(z)$ is analytic on X with no zeros in X . Then there exists an analytic function $h(z)$ on X such that

$$e^{h(z)} = g(z) \quad \text{for all } z \in X.$$

and we say that $h(z)$ is a branch of logarithm of $g(z)$ on X .



Note There is a mini definition in the above theorem: A branch of logarithm of $g(z)$ on X is an analytic function $h(z)$ on X such that $e^{h(z)} = g(z)$ for all $z \in X$.



Note We may also write $h(z) = \log g(z)$ to denote a branch of the logarithm of $g(z)$ on X . However, one must be careful when performing algebraic operations with the logarithm, since $\log g(z)$ is defined only up to a choice of branch. Nevertheless, many familiar properties of the logarithm still hold for a fixed branch. For example, the derivative of $h(z)$ is given by

$$h'(z) = \frac{g'(z)}{g(z)},$$

which is formally the same as the derivative of the real logarithm.

Indeed, differentiating both sides of the identity $e^{h(z)} = g(z)$ and applying the chain rule, we obtain

$$e^{h(z)} h'(z) = g'(z).$$

Since $e^{h(z)} = g(z)$, it follows that

$$h'(z) = \frac{g'(z)}{g(z)}.$$

Proof Since $g(z)$ has no zeros in X , we can define a function

$$f(z) := \int_{z_0}^z \frac{g'(w)}{g(w)} dw$$

for some fixed $z_0 \in X$. This integral is well-defined because X is simply connected, and $\frac{g'(w)}{g(w)}$ is analytic on X (as $g(w)$ has no zeros).

Now, consider $F(z) := g(z)e^{-f(z)}$. Notice that $F'(z) = g'(z)e^{-f(z)} - g(z)e^{-f(z)} \cdot \frac{g'(z)}{g(z)} = g'(z)e^{-f(z)} -$

$g'(z)e^{-f(z)} = 0$. Thus, $F(z)$ is constant on X . Evaluating at $z = z_0$, we have

$$F(z) = F(z_0) = g(z_0)e^{-f(z_0)} = g(z_0)e^0 = g(z_0) \implies g(z)e^{-f(z)} = F(z) = g(z_0). \implies g(z) = g(z_0)e^{f(z)}.$$

Therefore, we can set $h(z) := f(z) + c$ where $e^c = g(z_0)$, and we have $e^{h(z)} = g(z)$ for all $z \in X$.

 **Exercise 2.8** Consider the analytic function $f(z) = \text{Log}(1 - z)$ on the simply connected open set $X = \{z \in \mathbb{C} : |z| < 1\}$ where Log is the principal branch of logarithm defined on $\mathbb{C} \setminus (-\infty, 0]$. Show the $f(z)$ has the Taylor series expansion

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = -\sum_{n=1}^{\infty} \frac{z^n}{n} \quad \text{valid for } z \in X,$$

and hence deduce that when $|z| \leq \frac{1}{2}$, $|\text{Log}(1 - z)| \leq 2|z|$.

Solution Notice that for each $n \in \mathbb{N}$,

$$f^{(n)}(z) = \frac{d^n}{dz^n} \text{Log}(1 - z) = \frac{d^{n-1}}{dz^{n-1}} \left(-\frac{1}{1 - z} \right) = \frac{d^{n-2}}{dz^{n-2}} \left(-\frac{1}{(1 - z)^2} \right) = \cdots = -\frac{(n-1)!}{(1 - z)^n}, \quad \text{for } n \geq 1.$$

Thus,

$$f^{(n)}(0) = -(n-1)! \quad \text{for all } n \geq 1, \quad \text{and} \quad f(0) = \text{Log}(1 - 0) = \text{Log}(1) = 0.$$

Therefore, the Taylor series expansion of $f(z)$ at $z = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = f(0) + \sum_{n=1}^{\infty} \frac{-(n-1)!}{n!} z^n = -\sum_{n=1}^{\infty} \frac{z^n}{n}.$$

where the last equality holds since $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n$ converges for $|z| < 1$.

Next, we show that when $|z| \leq \frac{1}{2}$, we have $|\text{Log}(1 - z)| \leq 2|z|$. Since $|z| < \frac{1}{2}$, we have

$$|\text{Log}(1 - z)| = \left| -\sum_{n=1}^{\infty} \frac{z^n}{n} \right| \leq \sum_{n=1}^{\infty} \frac{|z|^n}{n} \leq \sum_{n=1}^{\infty} |z|^n = \frac{|z|}{1 - |z|} \leq \frac{|z|}{1 - \frac{1}{2}} = 2|z|.$$

Theorem 2.10 (Uniform Convergence of Logarithm of Infinite Product)

On a simply connected open set $X \subseteq \mathbb{C}$, the infinite product

$$f(z) = \prod_{n=1}^{\infty} (1 + u_n(z))$$

where $|u_n(z)| \leq M_n < 1$ for all $z \in X$ and $\sum_{n=1}^{\infty} M_n < \infty$, is uniformly convergent on X . Further more, the function

$$L(z) := \sum_{n=1}^{\infty} \text{Log}(1 + u_n(z)) \quad \text{where } \text{Log} \text{ is principal branch of logarithm,}$$

is convergent uniformly on X and satisfies

$$e^{L(z)} = f(z) \quad \text{for all } z \in X.$$

$L(z)$ is thus a branch of logarithm of $f(z)$ on X .

(i.e., “ $L(z) = \log f(z)$ ” and $L(z)$ is analytic on X , which is an application of the previous theorem 2.9.)

Proof $f(z)$ is clearly uniformly convergent on X by theorem 2.5 since $\sum_{n=1}^{\infty} M_n < \infty$ and $|u_n(z)| \leq M_n$ for all $z \in X$.

Next, we need to show that $L(z)$ is well-defined and uniformly convergent on X . For each $N \in \mathbb{N}$, we define

$$\widetilde{\log} P_N(z) := \sum_{n=1}^N \text{Log}(1 + u_n(z)), \quad \text{where } P_N(z) = \prod_{n=1}^N (1 + u_n(z)).$$

Since $|u_n(z)| \leq M_n < 1$ for all $z \in X$, we have $|1 + u_n(z)| \geq 1 - |u_n(z)| \geq 1 - M_n > 0$ for all $z \in X$. Thus, $\widetilde{\log} P_N(z)$ is well-defined and analytic on X . Further more, notice that

$$e^{\widetilde{\log} P_N(z)} = e^{\sum_{n=1}^N \text{Log}(1 + u_n(z))} = \prod_{n=1}^N e^{\text{Log}(1 + u_n(z))} = \prod_{n=1}^N (1 + u_n(z)) = P_N(z).$$

so that $\widetilde{\log} P_N(z)$ is a branch of logarithm of $P_N(z)$ on X . (i.e., $e^{\widetilde{\log} P_N(z)} = P_N(z)$ for all $z \in X$.) Since $|u_n(z)| \leq M_n$ and $\sum_{n=1}^{\infty} M_n < \infty$, $u_n(z) \leq \frac{1}{2}$ for all sufficiently large N_0 and all $z \in X$. Thus, by exercise 2.8, for all $N \geq N_0$ and

all $z \in X$, we have $|\operatorname{Log}(1 + u_n(z))| \leq 2|u_n(z)|$. Therefore, by the Weierstrass M-test 2.5, considering applying it to the series $\sum_{n=N_0}^{\infty} \operatorname{Log}(1 + u_n(z))$, we have

$$\widetilde{\log} P_N(z) \rightarrow L(z) := \sum_{n=1}^{\infty} \operatorname{Log}(1 + u_n(z)) \quad \text{uniformly on } X \text{ as } N \rightarrow \infty$$

and hence

$$P_N(z) = e^{\widetilde{\log} P_N(z)} \rightarrow e^{L(z)} \quad \text{uniformly on } X \text{ as } N \rightarrow \infty.$$

Recall that $P_N(z) \rightarrow f(z)$ uniformly on X as $N \rightarrow \infty$, we have

$$f(z) = e^{L(z)} \quad \text{for all } z \in X.$$

 **Exercise 2.9** Recall the Generalized Cauchy Integral Formula

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\Gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz,$$

where f is analytic inside and on a simple closed contour Γ in an open set $X \subseteq \mathbb{C}$ and z_0 is a point inside Γ .

Use this formula to show that if a sequence of analytic functions $\{f_n\}$ converges uniformly to a function f on an open set $X \subseteq \mathbb{C}$, then the sequence of derivatives $\{f'_n\}$ converges uniformly to f' on X . i.e., show that

$$f'_n \rightarrow f' \quad \text{uniformly on } X.$$

Solution Let $\{f_n\}$ be a sequence of analytic functions on an open set $X \subseteq \mathbb{C}$ that converges uniformly to a function f on X . We show that the sequence of derivatives $\{f'_n\}$ converges uniformly to f' on every compact subset of X .

Let $K \subset X$ be an arbitrary compact set. Since X is open and K is compact, there exists $\delta > 0$ such that

$$\operatorname{dist}(K, \partial X) \geq \delta.$$

Fix $z_0 \in K$ and let Γ be the positively oriented circle

$$\Gamma := \{z \in \mathbb{C} : |z - z_0| = r\}, \quad r := \delta/2.$$

Then $\Gamma \subset X$ for all $z_0 \in K$. By the Generalized Cauchy Integral Formula, we have

$$f'_n(z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f_n(z)}{(z - z_0)^2} dz, \quad f'(z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{(z - z_0)^2} dz.$$

Since $f_n \rightarrow f$ uniformly on X , for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and all $z \in X$,

$$|f_n(z) - f(z)| < \varepsilon.$$

Hence, for $n \geq N$, using the M-L inequality we obtain

$$\begin{aligned} |f'_n(z_0) - f'(z_0)| &= \left| \frac{1}{2\pi i} \int_{\Gamma} \frac{f_n(z) - f(z)}{(z - z_0)^2} dz \right| \\ &\leq \frac{1}{2\pi} \max_{z \in \Gamma} \left| \frac{f_n(z) - f(z)}{(z - z_0)^2} \right| \cdot \operatorname{length}(\Gamma). \end{aligned}$$

Since $|z - z_0| = r$ on Γ and $\operatorname{length}(\Gamma) = 2\pi r$, we have

$$|f'_n(z_0) - f'(z_0)| \leq \frac{1}{2\pi} \cdot \frac{\varepsilon}{r^2} \cdot 2\pi r = \frac{\varepsilon}{r}.$$

The constant $r = \delta/2$ is independent of $z_0 \in K$. Therefore,

$$\sup_{z_0 \in K} |f'_n(z_0) - f'(z_0)| \leq \frac{\varepsilon}{r}, \quad \text{for all } n \geq N.$$

This shows that $f'_n \rightarrow f'$ uniformly on K . Since $K \subset X$ was arbitrary, we conclude that

$$f'_n \rightarrow f' \quad \text{uniformly on compact subsets of } X.$$

Corollary 2.3

Let $\zeta(z)$ be zeta function defined as in corollary 2.2, we have

$$\frac{\zeta'(z)}{\zeta(z)} = - \sum_{p \in \mathbb{P}} \frac{\log p}{p^z - 1} \quad \text{for } \Re(z) > 1.$$





Note It's a very important formula in the proof of the prime number theorem (PNT) later.

Proof For the zeta function $\zeta(z)$, by corollary 2.2, we have

$$\zeta(z) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}} = \prod_{p \in \mathbb{P}} (1 + u_p(z)) \quad \text{where } u_p(z) = \frac{1}{p^z - 1}.$$

Consider a simply connected open set $X_\varepsilon = \{z \in \mathbb{C} : \Re(z) > 1 + \varepsilon\}$ for some $\varepsilon > 0$, we have

$$|u_p(z)| = \left| \frac{1}{p^z - 1} \right| \leq 2|p^{-z}| = 2p^{-\Re(z)} \leq 2p^{-(1+\varepsilon)} \quad \text{for all } z \in X_\varepsilon.$$

Let $M_p = 2p^{-(1+\varepsilon)}$, we have $\sum_{p \in \mathbb{P}} M_p < \infty$ (Comparison test with the p-series test.) and $M_p < 1$ for all primes p . Thus, by theorem 2.10, we have

$$L(z) := \sum_{p \in \mathbb{P}} \text{Log}(1 + u_p(z)) = \sum_{p \in \mathbb{P}} \text{Log} \left(\frac{1}{1 - p^{-z}} \right) = - \sum_{p \in \mathbb{P}} \text{Log}(1 - p^{-z})$$

is uniformly convergent on X_ε and satisfies

$$e^{L(z)} = \zeta(z) \quad \text{for all } z \in X_\varepsilon.$$

Then, by exercise 2.9, considering the sequence of partial sums of $L(z)$, denoted by $S_N(z) = - \sum_{p \leq N} \text{Log}(1 - p^{-z})$, we have

$$S'_N(z) = - \sum_{p \leq N} \frac{p^{-z} \log p}{1 - p^{-z}} = - \sum_{p \leq N} \frac{\log p}{p^z - 1} \rightarrow L'(z) \quad \text{uniformly on } X_\varepsilon \text{ as } N \rightarrow \infty.$$

As $e^{L(z)} = \zeta(z)$ for all $z \in X_\varepsilon$, by the chain rule, we have

$$L'(z)e^{L(z)} = L'(z)\zeta(z) = \zeta'(z) \implies L'(z) = \frac{\zeta'(z)}{\zeta(z)}.$$

Therefore, for all z such that $\Re(z) > 1$, we can choose $\varepsilon > 0$ such that $z \in X_\varepsilon$, and hence

$$\frac{\zeta'(z)}{\zeta(z)} = L'(z) = - \sum_{p \in \mathbb{P}} \frac{\log p}{p^z - 1} \quad \text{for } z \in X_\varepsilon.$$

Corollary 2.4

The Dirichlet L -function $L(z, \chi)$ for a Dirichlet character χ satisfies

$$L(z, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-z}} \quad \text{for } \Re(z) > 1,$$

(although we have not defined Dirichlet characters yet, and not proved the Euler product formula for Dirichlet L -functions, but we will do so later, just take it as given for now).

Then we have

$$\log L(z, \chi) := \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{1}{k} \chi(p^k) p^{-kz} \quad \text{for } \Re(z) > 1,$$

is a branch of logarithm of $L(z, \chi)$ on the open set $\{z \in \mathbb{C} : \Re(z) > 1\}$, i.e.,

$$e^{\log L(z, \chi)} = L(z, \chi) \quad \text{for } \Re(z) > 1.$$



Note Be careful that here $\log L(z, \chi)$ is defined as a branch of logarithm of $L(z, \chi)$ on the open set $\{z \in \mathbb{C} : \Re(z) > 1\}$, not the natural logarithm of real numbers.

Proof For the Dirichlet L -function $L(z, \chi)$, we have

$$L(z, \chi) = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-z}} = \prod_{p \in \mathbb{P}} (1 + u_p(z)) \quad \text{where } u_p(z) = \frac{\chi(p)}{p^z - \chi(p)}.$$

Consider a simply connected open set $X_\varepsilon = \{z \in \mathbb{C} : \Re(z) > 1 + \varepsilon\}$ for some $\varepsilon > 0$, we have

$$|u_p(z)| = \left| \frac{\chi(p)}{p^z - \chi(p)} \right| \leq 2|\chi(p)||p^{-z}| = 2p^{-\Re(z)} \leq 2p^{-(1+\varepsilon)} \quad \text{for all } z \in X_\varepsilon.$$

Let $M_p = 2p^{-(1+\varepsilon)}$, we have $\sum_{p \in \mathbb{P}} M_p < \infty$ (Comparison test with the p-series test.) and $M_p < 1$ for all primes p .

Thus, by theorem 2.10, we have

$$L(z) := \sum_{p \in \mathbb{P}} \text{Log}(1 + u_p(z)) = \sum_{p \in \mathbb{P}} \text{Log} \left(\frac{1}{1 - \chi(p)p^{-z}} \right) = - \sum_{p \in \mathbb{P}} \text{Log}(1 - \chi(p)p^{-z})$$

is uniformly convergent on X_ε and satisfies

$$e^{L(z)} = L(z, \chi) \quad \text{for all } z \in X_\varepsilon.$$

Next, we expand $\text{Log}(1 - \chi(p)p^{-z})$ using the Taylor series expansion from exercise 2.8, $\text{Log}(1 - z) = -\sum_{n=1}^{\infty} \frac{z^n}{n}$ for $|z| < 1$, we have

$$\text{Log}(1 - \chi(p)p^{-z}) = - \sum_{k=1}^{\infty} \frac{(\chi(p)p^{-z})^k}{k} = - \sum_{k=1}^{\infty} \frac{1}{k} \chi(p^k) p^{-kz}.$$

Therefore,

$$L(z) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{1}{k} \chi(p^k) p^{-kz} \quad \text{for } z \in X_\varepsilon.$$

Finally, for all z such that $\Re(z) > 1$, we can choose $\varepsilon > 0$ such that $z \in X_\varepsilon$, and hence

$$e^{\log L(z, \chi)} = L(z, \chi) \quad \text{for } \Re(z) > 1.$$

Exercises

1. Let $\{a_n\}$ be a sequence of complex number such that the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit P . Show that

- (a) The infinite product $\prod_{n=1}^{\infty} a_n^{-1}$ converges and

$$\prod_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{P}$$

- (b) Hence conclude that for $\Re(z) > 1$,

$$\frac{1}{\zeta(z)} = \prod_{p \text{ prime}} (1 - p^{-z})$$

Solution

(a)

2. The Möbius function $\mu : \mathbb{N} \rightarrow \mathbb{C}$ is defined as follows:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 \cdots p_k \text{ is a product of } k \text{ distinct primes,} \\ 0 & \text{otherwise,} \end{cases}$$

Show that

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

(Hint: The case $n = 1$ is easy. For $n > 1$, consider the prime factorization of $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ so that the divisors d of n are of the form $d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}$ where $0 \leq \beta_i \leq \alpha_i$ for all $1 \leq i \leq r$. The only divisors d which give a non-zero contribution to the sum are those d such that $\beta_i \in \{0, 1\}$ for all $1 \leq i \leq r$. So that

$$\sum_{d|n} \mu(d) = \mu(1) + (\mu(p_1) + \mu(p_2) + \cdots + \mu(p_r)) + (\mu(p_1 p_2) + \mu(p_1 p_3) + \cdots) + \cdots + \mu(p_1 p_2 \cdots p_r).$$

Now, use the binomial theorem, compare this to $0 = (1 - 1)^r = \sum_{k=0}^r \binom{r}{k} (-1)^k$.

Solution

3. Let $f(n) = \mu(n)$ denote the Möbius function as defined in the previous exercise and let $g(n) = 1$ for all $n \in \mathbb{N}$. Show that

(a)

$$f * g := \sum_{d|n} f(d)g\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

(b) Hence conclude that

$$\frac{1}{\zeta(z)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^z} \quad \text{for } \operatorname{Re}(z) > 1.$$

Solution

(a)

4. Let $d(n) := \#$ of divisors of n , and let $f(n) = g(n) = 1$ for all $n \in \mathbb{N}$. Show that

(a)

$$f * g := \sum_{d|n} f(d)g\left(\frac{n}{d}\right) = d(n).$$

(b) Hence conclude that

$$\zeta^2(z) = \sum_{n=1}^{\infty} \frac{d(n)}{n^z} \quad \text{for } \operatorname{Re}(z) > 1.$$

Solution

(a)

5. Let $\phi(n) := \#$ of integers $1 \leq k \leq n$ such that $\gcd(k, n) = 1$ denote the Euler totient function. Show that

(a) For $n \in \mathbb{N}$,

$$\sum_{d|n} \phi(d) = n.$$

(b) Let $f(n) = \phi(n)$ and $g(n) = 1$ for all $n \in \mathbb{N}$. Show that

$$f * g := \sum_{d|n} f(d)g\left(\frac{n}{d}\right) = n.$$

(c) Hence conclude that

$$\frac{\zeta(z-1)}{\zeta(z)} = \sum_{n=1}^{\infty} \frac{\phi(n)}{n^z} \quad \text{for } \operatorname{Re}(z) > 2.$$

Solution

(a)

6. Let $f : \mathbb{N} \rightarrow \mathbb{C}$, then

$$F(z) := \sum_{n=1}^{\infty} \frac{f(n)}{n^z}$$

is called the *Dirichlet series* associated to f . You may find that the previous exercises 4. and 5. give some examples of Dirichlet series and satisfies certain properties, which can be summarized in the following

Theorem: Suppose $F(z) = \sum_{n=1}^{\infty} \frac{f(n)}{n^z}$ and $G(z) = \sum_{n=1}^{\infty} \frac{g(n)}{n^z}$ are two Dirichlet series that converge for some $z \in \mathbb{C}$. Then the product $F(z)G(z)$

$$F(z)G(z) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^z}$$

is also a Dirichlet series, where

$$(f * g)(n) := \sum_{d|n} f(d)g\left(\frac{n}{d}\right).$$

is called the *Dirichlet convolution* of f and g , which is a finite sum for each $n \in \mathbb{N}$.

Now, with the following steps, prove the above theorem.

- (a) Suppose the Dirichlet series $F(z) = \sum_{n=1}^{\infty} \frac{f(n)}{n^z}$ and $G(z) = \sum_{n=1}^{\infty} \frac{g(n)}{n^z}$ converge for some $z \in \mathbb{C}$. For

$N > 1$, write

$$\begin{aligned} \sum_{n=1}^N \frac{f(n)}{n^z} \cdot \sum_{m=1}^N \frac{g(m)}{m^z} &= \sum_{k=1}^{N^2} \frac{1}{k^z} \left[\sum_{\substack{m=1 \\ mn=k}}^N \sum_{n=1}^N f(n)g(m) \right] = \sum_{k=1}^{N^2} \frac{1}{k^z} \left[\sum_{\substack{n=1 \\ n|k}}^N f(n)g\left(\frac{k}{n}\right) \right] \\ &= \sum_{k=1}^{N^2} \frac{(f * g)(k)}{k^z} - \sum_{k=N+1}^{N^2} \frac{1}{k^z} \left[\sum_{\substack{n>N \\ n|k}} f(n)g\left(\frac{k}{n}\right) \right]. \end{aligned}$$

Let

$$A_N := \sum_{k=N+1}^{N^2} \frac{1}{k^z} \left[\sum_{\substack{n>N \\ n|k}} f(n)g\left(\frac{k}{n}\right) \right].$$

Show that

$$A_N = \sum_{\substack{n=N+1 \\ N+1 \leq nm \leq N^2}}^{\infty} \sum_{m=1}^N \frac{f(n)}{n^z} \cdot \frac{g(m)}{m^z} \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

(b) Hence conclude that

$$F(z)G(z) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^z}.$$

so that we have proved the theorem.

Solution

(a)

Chapter 3 Dirichlet Characters

3.1 Definition and Basic Properties

Definition 3.1 (Dirichlet Character)

A function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ is called a Dirichlet character modulo q if the following conditions hold:

- (i) χ is completely multiplicative, i.e., for all $m, n \in \mathbb{Z}$, $\chi(mn) = \chi(m)\chi(n)$.
- (ii) χ is periodic with period q , i.e., for all $n \in \mathbb{Z}$, $\chi(n + q) = \chi(n)$.
- (iii) $\chi(n) \neq 0$ if and only if $\gcd(n, q) = 1$.



 **Note** By the definition of Dirichlet character, we have $\chi(1) = 1$ since $\chi(1) = \chi(1 \cdot 1) = \chi(1)\chi(1)$ and $\chi(1) \neq 0$ as $\gcd(1, q) = 1$. So $\chi(1) = 1$ for any Dirichlet character χ .

 **Exercise 3.1** For $m, n \in \mathbb{Z}$ and q a positive integer, show that:

$$\gcd(mn, q) = 1 \iff \gcd(m, q) = 1 \text{ and } \gcd(n, q) = 1.$$

Solution ($\Rightarrow$) Suppose that $\gcd(mn, q) = 1$. Then there are integers a, b such that

$$a(mn) + bq = 1.$$

Thus, any common divisor of m and q must divide the right-hand side, which is 1. Therefore, $\gcd(m, q) = 1$. Similarly, any common divisor of n and q must also divide 1, so $\gcd(n, q) = 1$.

($\Leftarrow$) Conversely, suppose that $\gcd(m, q) = 1$ and $\gcd(n, q) = 1$. Then there exist integers a_1, b_1 such that

$$a_1m + b_1q = 1,$$

and integers a_2, b_2 such that

$$a_2n + b_2q = 1.$$

Multiplying these two equations together, we get

$$(a_1m + b_1q)(a_2n + b_2q) = 1.$$

Expanding the left-hand side, we have

$$a_1a_2mn + (a_1b_2m + a_2b_1n + b_1b_2q)q = 1.$$

This shows that any common divisor of mn and q must divide the right-hand side, which is 1. Therefore, $\gcd(mn, q) = 1$.

Proposition 3.1

Let χ be a Dirichlet character modulo q . Consider a function $\tilde{\chi} : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$ defined by

$$\tilde{\chi}([n]) := \chi(n)$$

where $[n]$ is the equivalence class of n in $\mathbb{Z}/q\mathbb{Z}$.

Then, $\tilde{\chi}$ is well-defined on $\mathbb{Z}/q\mathbb{Z}$.



Remark Recall that $\mathbb{Z}/q\mathbb{Z}$ is a finite cyclic group under addition modulo q , and further more, if we add the multiplication operation modulo q , then it becomes a finite ring.

One may consider $\tilde{\chi}$ as a function defined on the finite ring $\mathbb{Z}/q\mathbb{Z}$ but note that $\tilde{\chi}$ is not a ring homomorphism since it is not additive in general, i.e., $\tilde{\chi}([m] + [n]) \neq \tilde{\chi}([m]) + \tilde{\chi}([n])$ in general.

 **Note** After this proposition, we will often identify χ with $\tilde{\chi}$ and simply write $\chi : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$. Hopefully this abuse of notation will not cause any confusion.

 **Exercise 3.2** Prove the above proposition.

Solution To show that $\tilde{\chi}$ is well-defined, we need to verify that if $[n] = [m]$ in $\mathbb{Z}/q\mathbb{Z}$, then $\tilde{\chi}([n]) = \tilde{\chi}([m])$.

By the definition of equivalence classes in $\mathbb{Z}/q\mathbb{Z}$, we have $[n] = [m]$ if and only if $n \equiv m \pmod{q}$. This means that

there exists an integer k such that

$$n = m + kq.$$

Since χ is periodic with period q , we have

$$\chi(n) = \chi(m + kq) = \chi(m).$$

Therefore,

$$\tilde{\chi}([n]) = \chi(n) = \chi(m) = \tilde{\chi}([m]).$$

This shows that $\tilde{\chi}$ is well-defined on $\mathbb{Z}/q\mathbb{Z}$.

Definition 3.2 (Group of Units of $\mathbb{Z}/q\mathbb{Z}$)

Let $\mathbb{Z}/q\mathbb{Z}$ be the finite **ring** of integers modulo q . The group of units of $\mathbb{Z}/q\mathbb{Z}$, denoted by G_q , is the set of all elements in $\mathbb{Z}/q\mathbb{Z}$ that have a multiplicative inverse. We regard G_q as a finite **group** under multiplication modulo q .



Example 3.1 Here, we review some properties in elementary number theory to introduce the following lemma.

Let $I_{m,n} = \{mx + ny : x, y \in \mathbb{Z}\} = \langle m, n \rangle$ be the ideal generated by m and n in the ring of integers $\mathbb{Z}$. Then it is a well-known result that

$$I_{m,n} = d\mathbb{Z} \quad \text{where } d = \gcd(m, n)$$

(i.e. $\langle m, n \rangle = \langle \gcd(m, n) \rangle$). If you are familiar with the concept of a principal ideal domain (PID), then this result follows directly from the fact that $\mathbb{Z}$ is a PID. Here, however, we give a fundamental proof without using the concept of PID. Our goal is first to show that $I_{m,n} = d\mathbb{Z}$ for some $d \in \mathbb{Z}$, and then to show that this d is indeed $\gcd(m, n)$.

First, we show that $I_{m,n} \subseteq d\mathbb{Z}$ for some $d \in \mathbb{Z}$. Let d be the smallest positive integer in $I_{m,n}$, so there exist integers x_0, y_0 such that $d = mx_0 + ny_0$. For any element $a \in I_{m,n}$, we have $a = mx_1 + ny_1$ for some integers x_1, y_1 . By the division algorithm, there exist unique integers q, r such that $a = qd + r$ with $0 \leq r < d$. Thus,

$$r = a - qd = (mx_1 + ny_1) - q(mx_0 + ny_0) = m(x_1 - qx_0) + n(y_1 - qy_0) \in I_{m,n}.$$

Since d is the smallest positive integer in $I_{m,n}$, we must have $r = 0$. Therefore, $a = qd \in d\mathbb{Z}$, and hence $I_{m,n} \subseteq d\mathbb{Z}$.

Now we show that $d\mathbb{Z} \subseteq I_{m,n}$. Since $d = mx_0 + ny_0$, for any integer k we have

$$kd = k(mx_0 + ny_0) = m(kx_0) + n(ky_0) \in I_{m,n}.$$

Thus, $d\mathbb{Z} \subseteq I_{m,n}$, and hence $I_{m,n} = d\mathbb{Z}$.

Furthermore, since both m and n belong to $I_{m,n}$, d divides both m and n . Thus, d is a common divisor of m and n . If there exists a common divisor d' of m and n such that $d' > d$, then d' divides every element of $I_{m,n}$, which contradicts the definition of d as the smallest positive integer in $I_{m,n}$. Therefore, $d = \gcd(m, n)$.

Lemma 3.1 (Characterization of Group of Units of $\mathbb{Z}/q\mathbb{Z}$)

The group of units of $\mathbb{Z}/q\mathbb{Z}$ is given by

$$G_q = \{[n] \in \mathbb{Z}/q\mathbb{Z} : \gcd(n, q) = 1\}.$$

Sometimes, we also write $G_q = \{0 \leq r \leq q-1 : \gcd(r, q) = 1\}$ by using r to denote the representative of the equivalence class $[r]$ in $\mathbb{Z}/q\mathbb{Z}$.

Furthermore, by definition of Euler's totient function $\phi(q)$, we have

$$|G_q| = \phi(q).$$



Proof For a given $[n] \in \mathbb{Z}/q\mathbb{Z}$, we have $[n] \in G_q$ if and only if there exists $[m] \in \mathbb{Z}/q\mathbb{Z}$ such that

$$[n][m] = [1] \implies [nm] = [1] \implies nm \equiv 1 \pmod{q}.$$

This is equivalent to saying that for which $n \in \mathbb{Z}$ the linear congruence equation $nx \equiv 1 \pmod{q}$ has a solution. And this is equivalent to solve the linear Diophantine equation

$$nx - qy = 1$$

for integers x, y . By example 3.1, we know that $nx - qy \in \gcd(n, q)\mathbb{Z}$ for all integers x, y . Thus, the above equation has integer solutions only if $\gcd(n, q) = 1$. (Otherwise, $1 \notin \gcd(n, q)\mathbb{Z}$.) Thus, we have shown that

$$G_q \subseteq \{[n] \in \mathbb{Z}/q\mathbb{Z} : \gcd(n, q) = 1\}.$$

Conversely, suppose that $\gcd(n, q) = 1$. Then $1 \in I_{n, q}$ by example 3.1, so there exist integers x_0, y_0 such that $nx_0 + qy_0 = 1$. The equation $nx \equiv 1 \pmod{q}$ has a solution $x \equiv x_0 \pmod{q}$. Thus, there exists $[m] \in \mathbb{Z}/q\mathbb{Z}$ such that $[n][m] = [1]$, and hence $[n] \in G_q$.

Corollary 3.1 (Euler's Theorem)

For any integer n such that $\gcd(n, q) = 1$, we have

$$n^{\phi(q)} \equiv 1 \pmod{q}.$$

Proof Since $\gcd(n, q) = 1$, we have $[n] \in G_q$. By Corollary of Lagrange's Theorem (If $g \in G$, G is a finite group, then $g^{|G|} = e$ is the identity element of G), we have

$$[n]^{|G_q|} = [1] \implies [n]^{\phi(q)} = [1] \implies n^{\phi(q)} \equiv 1 \pmod{q}.$$

Corollary 3.2 (Property of Dirichlet Characters from Euler's Theorem)

Let χ be a Dirichlet character modulo q . Then for any $n \in \mathbb{Z}$ such that $\gcd(n, q) = 1$, we have

$$\chi(n)^{\phi(q)} = 1$$

Proof Direct from corollary 3.1 and the fact that χ is completely multiplicative, we have

$$\chi(n)^{\phi(q)} = \chi(n^{\phi(q)}) = \chi(1) = 1.$$

Corollary 3.3 (Dirichlet Characters are Roots of Unity)

Let χ be a Dirichlet character modulo q . Then we have

$$\chi(n) = \begin{cases} 0 & \text{if } \gcd(n, q) \neq 1, \\ e^{\frac{2\pi i \cdot k_q(n)}{\phi(q)}} & \text{for some } k_q(n) \in \{0, 1, 2, \dots, \phi(q) - 1\} \text{ if } \gcd(n, q) = 1. \end{cases}$$

Proof Direct from corollary 3.2, we have $\chi(n)^{\phi(q)} = 1$ for all n such that $\gcd(n, q) = 1$. Thus, $\chi(n)$ is a $\phi(q)$ -th root of unity. The $\phi(q)$ -th roots of unity are given by

$$e^{2\pi i k / \phi(q)} \quad \text{for } k = 0, 1, 2, \dots, \phi(q) - 1.$$

and for n such that $\gcd(n, q) \neq 1$, we have $\chi(n) = 0$ by definition of Dirichlet characters.

Remark Clearly, $k = k_q(n)$ may depend on both q and n . Thus, understanding a Dirichlet character $\chi \pmod{q}$ boils down to understanding the map

$$n \mapsto k_q(n) \quad \text{for } n \text{ such that } \gcd(n, q) = 1.$$

3.2 Characters in Group Theory

Definition 3.3 (Character of a Finite Abelian Group)

Let G be a finite abelian group. A character of G is a group homomorphism $\chi : G \rightarrow \mathbb{C} \setminus \{0\}$, where $\mathbb{C} \setminus \{0\}$ is regarded as a group under multiplication.



Note Recall that a group homomorphism $\varphi : G \rightarrow H$ between two groups G and H is a map such that for all $g_1, g_2 \in G$,

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2).$$

Proposition 3.2 (Characters have Modulus 1)

Let χ be a character of a finite abelian group G . Then, $|\chi(g)| = 1$, for all $g \in G$.

 **Exercise 3.3** Prove the above proposition.

Solution (Recall that for a homomorphism $\varphi : G \rightarrow H$, we have $\varphi(e_G) = e_H$, where e_G and e_H are the identity elements of G and H respectively.) Thus, for any $g \in G$ with order d , we have

$$1 = \chi(e_G) = \chi(g^{|G|}) = (\chi(g))^{|G|} \implies (\chi(g))^{|G|} = 1 \implies |\chi(g)| = 1.$$

Definition 3.4 (Principal Character)

Let G be a finite abelian group. The character $\chi_0 : G \rightarrow \mathbb{C} \setminus \{0\}$ defined by

$$\chi_0(g) = 1 \quad \text{for all } g \in G$$

is called the Principal Character of G .

Definition 3.5 (Group of Characters)

Let G be a finite abelian group. The set of all characters of G is denoted by $\widehat{G}$, i.e.,

$$\widehat{G} := \{\chi : \chi \text{ is a character of } G\}.$$

 **Exercise 3.4** Let G be a finite abelian group and $g^* \in G$ be any element. Show that $H := \{gg^* : g \in G\}$ has the same cardinality as G . (i.e. Elements of G and H are same.)

Solution We only need to show that the map $f : G \rightarrow G$ defined by $f(g) = gg^*$ is bijective. Clearly it is surjective so we only need to show that it is injective.

Suppose that $f(g_1) = f(g_2)$ for some $g_1, g_2 \in G$. Then, we have

$$g_1g^* = g_2g^* \implies g_1g^*(g^*)^{-1} = g_2g^*(g^*)^{-1} \implies g_1 = g_2$$

since G is a group and hence every element has a multiplicative inverse. Thus, f is injective.

Theorem 3.1 (Orthogonality Relations of Characters)

For a finite abelian group G , let $\chi_1, \chi_2 \in \widehat{G}$, define the inner product of χ_1 and χ_2 by

$$\langle \chi_1, \chi_2 \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_1(g) \overline{\chi_2(g)}.$$

Then, we have

$$\langle \chi_1, \chi_2 \rangle = \begin{cases} 1 & \text{if } \chi_1 = \chi_2, \\ 0 & \text{if } \chi_1 \neq \chi_2. \end{cases}$$

Proof We omitted the inner product proof of $\langle \chi_1, \chi_2 \rangle$ here. Just show the last part that $\langle \chi_1, \chi_2 \rangle = \delta_{\chi_1, \chi_2}$ where δ is the Kronecker delta.

If $\chi_1 = \chi_2$, $\chi_1(g) \overline{\chi_2(g)} = |\chi_1(g)|^2 = 1$ for all $g \in G$ by proposition 3.2. Thus, sum of them is $|G|$ and hence $\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \cdot |G| = 1$.

If $\chi_1 \neq \chi_2$, Let $\psi(g) := \chi_1(g) \overline{\chi_2(g)}$ is also a character of G . If ψ is the principal character, then $\psi(g) = 1$ for all $g \in G$, which implies that for all $g \in G$,

$$\chi_1(g) \overline{\chi_2(g)} = 1 \implies \chi_1(g) \overline{\chi_2(g)} \chi_2(g) = \chi_2(g) \implies \chi_1(g) = \chi_2(g)$$

contradicting the assumption that $\chi_1 \neq \chi_2$. Thus, ψ is not the principal character, so there exists some $g_0 \in G$ such that $\psi(g_0) \neq 1$. Then, by exercise 3.4, we have

$$\sum_{g \in G} \psi(g) = \sum_{g \in G} \psi(gg_0) = \psi(g_0) \sum_{g \in G} \psi(g).$$

Since $\psi(g_0) \neq 1$, we have $\sum_{g \in G} \psi(g) = 0$. Therefore, $\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \cdot 0 = 0$.

 **Exercise 3.5** Let G be a finite set and consider the set $\mathbb{C}^G := \{f : G \rightarrow \mathbb{C} | f \text{ is any function}\}$. Show that $\mathbb{C}^G$ is a vector

space over the field $\mathbb{C}$ with pointwise addition and scalar multiplication, i.e., for $f_1, f_2 \in \mathbb{C}^G$ and $\alpha \in \mathbb{C}$,

$$(f_1 + f_2)(g) := f_1(g) + f_2(g), \quad (\alpha f_1)(g) := \alpha \cdot f_1(g) \quad \text{for all } g \in G.$$

Solution We need to verify the vector space conditions for $\mathbb{C}^G$.

(i) (Closure under addition) For any $f_1, f_2 \in \mathbb{C}^G$, $(f_1 + f_2)(g) = f_1(g) + f_2(g) \in \mathbb{C}$ for all $g \in G$. Thus, $f_1 + f_2 \in \mathbb{C}^G$.

(ii) (Closure under scalar multiplication) For any $f_1 \in \mathbb{C}^G$ and $\alpha \in \mathbb{C}$, $(\alpha f_1)(g) = \alpha \cdot f_1(g) \in \mathbb{C}$ for all $g \in G$. Thus, $\alpha f_1 \in \mathbb{C}^G$.

(iii) (Associativity of addition) For any $f_1, f_2, f_3 \in \mathbb{C}^G$, we have

$$((f_1 + f_2) + f_3)(g) = (f_1 + f_2)(g) + f_3(g) = f_1(g) + f_2(g) + f_3(g) = f_1(g) + (f_2 + f_3)(g) = (f_1 + (f_2 + f_3))(g).$$

Thus, $(f_1 + f_2) + f_3 = f_1 + (f_2 + f_3)$.

(iv) (Commutativity of addition) For any $f_1, f_2 \in \mathbb{C}^G$, we have

$$(f_1 + f_2)(g) = f_1(g) + f_2(g) = f_2(g) + f_1(g) = (f_2 + f_1)(g).$$

Thus, $f_1 + f_2 = f_2 + f_1$.

(v) (Existence of additive identity) Define the zero function $0 : G \rightarrow \mathbb{C}$ by $0(g) = 0$ for all $g \in G$. Then, for any $f \in \mathbb{C}^G$, we have

$$(f + 0)(g) = f(g) + 0(g) = f(g).$$

Thus, $f + 0 = f$.

(vi) (Existence of additive inverses) For any $f \in \mathbb{C}^G$, define $-f : G \rightarrow \mathbb{C}$ by $(-f)(g) = -f(g)$ for all $g \in G$. Then, we have

$$(f + (-f))(g) = f(g) + (-f)(g) = f(g) - f(g) = 0.$$

Thus, $f + (-f) = 0$.

(vii) (Distributivity of scalar multiplication over vector addition) For any $\alpha \in \mathbb{C}$ and $f_1, f_2 \in \mathbb{C}^G$, we have

$$(\alpha(f_1 + f_2))(g) = \alpha(f_1 + f_2)(g) = \alpha(f_1(g) + f_2(g)) = \alpha f_1(g) + \alpha f_2(g) = (\alpha f_1 + \alpha f_2)(g).$$

Thus, $\alpha(f_1 + f_2) = \alpha f_1 + \alpha f_2$.

(viii) (Distributivity of scalar multiplication over field addition) For any $\alpha, \beta \in \mathbb{C}$ and $f \in \mathbb{C}^G$, we have

$$((\alpha + \beta)f)(g) = (\alpha + \beta)f(g) = \alpha f(g) + \beta f(g) = (\alpha f + \beta f)(g).$$

Thus, $(\alpha + \beta)f = \alpha f + \beta f$.

(ix) (Compatibility of scalar multiplication with field multiplication) For any $\alpha, \beta \in \mathbb{C}$ and $f \in \mathbb{C}^G$, we have

$$((\alpha\beta)f)(g) = (\alpha\beta)f(g) = \alpha(\beta f(g)) = (\alpha(\beta f))(g).$$

Thus, $(\alpha\beta)f = \alpha(\beta f)$.

(x) (Existence of multiplicative identity) For any $f \in \mathbb{C}^G$, we have

$$(1f)(g) = 1 \cdot f(g) = f(g).$$

Thus, $1f = f$.

Therefore, $\mathbb{C}^G$ is a vector space over the field $\mathbb{C}$.

 **Exercise 3.6** Show that the vector space $\mathbb{C}^G$ over the field $\mathbb{C}$ which is defined in exercise 3.5 has dimension $|G|$.

Solution Consider the set of functions $\{e_g : g \in G\}$ defined by

$$e_g(h) = \begin{cases} 1 & \text{if } h = g, \\ 0 & \text{if } h \neq g. \end{cases}$$

for all $h \in G$. We will show that this set forms a basis for the vector space $\mathbb{C}^G$.

First, we show that the set $\{e_g : g \in G\}$ spans $\mathbb{C}^G$. For any function $f \in \mathbb{C}^G$, consider the linear combination

$$g = \sum_{g \in G} f(g)e_g.$$

Evaluating this linear combination at any $h \in G$, we have

$$g(h) = \sum_{g \in G} f(g)e_g(h) = f(h)e_h(h) + \sum_{\substack{g \in G \\ g \neq h}} f(g)e_g(h) = f(h) \cdot 1 + \sum_{\substack{g \in G \\ g \neq h}} f(g) \cdot 0 = f(h).$$

Thus, $g(h) = f(h)$ for all $h \in G$, which implies that $g = f$. This shows that any function in $\mathbb{C}^G$ can be written as a linear combination of the functions e_g , so the set spans $\mathbb{C}^G$.

Next, we show that the set $\{e_g : g \in G\}$ is linearly independent. Suppose that there exist scalars $\alpha_g \in \mathbb{C}$ such that

$$\sum_{g \in G} \alpha_g e_g = 0,$$

where 0 is the zero function. Evaluating this equation at each $h \in G$, we have

$$\sum_{g \in G} \alpha_g e_g(h) = 0.$$

Since $e_g(h) = 1$ if $h = g$ and 0 otherwise, this simplifies to

$$\alpha_h = 0.$$

Since this holds for all $h \in G$, we conclude that all coefficients α_g must be zero. Therefore, the set $\{e_g : g \in G\}$ is linearly independent.

Since the set $\{e_g : g \in G\}$ spans $\mathbb{C}^G$ and is linearly independent, it forms a basis for $\mathbb{C}^G$. The number of elements in this basis is equal to the cardinality of G , which is $|G|$. Therefore, the dimension of the vector space $\mathbb{C}^G$ is $|G|$.

Corollary 3.4 (Number of Characters is at most the Order of the Group)

For a finite abelian group G , and its group of characters $\widehat{G}$, we have

$$|\widehat{G}| \leq |G|.$$

Proof We can view $\widehat{G}$ as a subset of the vector space $\mathbb{C}^G$ over the field $\mathbb{C}$ defined in exercise 3.5. By theorem 3.1, the characters in $\widehat{G}$ are orthonormal with respect to the inner product defined there. Thus, the characters in $\widehat{G}$ are linearly independent in the vector space $\mathbb{C}^G$. This is one of the standard results in linear algebra: Any orthonormal set in an inner product space is linearly independent and here is the proof: Suppose that there exist scalars $\alpha_i \in \mathbb{C}$ such that $\sum_{\chi_i \in \widehat{G}} \alpha_i \chi_i = 0$, then

$$\left\langle \sum_{\chi_i \in \widehat{G}} \alpha_i \chi_i, \chi_j \right\rangle = 0 \implies \sum_{\chi_i \in \widehat{G}} \alpha_i \langle \chi_i, \chi_j \rangle = 0 \implies \sum_{\chi_i \in \widehat{G}} \alpha_i \delta_{\chi_i, \chi_j} = 0 \implies \alpha_j = 0.$$

Therefore, by exercise 3.6, we have

$$|\widehat{G}| = \dim(\text{span}(\widehat{G})) \leq \dim(\mathbb{C}^G) = |G|.$$

Theorem 3.2 (Number of Characters of Cyclic Group and Dirichlet Characters)

If $G = \mathbb{Z}/q\mathbb{Z}$, then there are exactly $|G| = q$ characters of G , i.e., $|\widehat{G}| = q$ and they are given by

$$\chi_k([n]) = e^{\frac{2\pi i k n}{q}} \quad \text{for } k = 0, 1, 2, \dots, q-1.$$

Furthermore, $\widehat{G} \cong G$.

Proof First, we show that $\chi_k([n])$ is indeed a character of G . For any $[m], [n] \in G$, we have

$$\chi_k([m] + [n]) = \chi_k([m+n]) = e^{\frac{2\pi i k (m+n)}{q}} = e^{\frac{2\pi i k m}{q}} \cdot e^{\frac{2\pi i k n}{q}} = \chi_k([m]) \cdot \chi_k([n]).$$

Thus, χ_k is a group homomorphism from G to $\mathbb{C} \setminus \{0\}$, so it is a character of G .

Next, we show that the characters χ_k for $k = 0, 1, 2, \dots, q-1$ are distinct. Since $\chi_k = \chi_{k+q}$ for all integers k , it suffices to show that $\chi_k \neq \chi_{k'}$ for $k, k' \in \{0, 1, 2, \dots, q-1\}$ with $k \neq k'$. Suppose that $\chi_k = \chi_{k'}$ for some $k, k' \in \{0, 1, 2, \dots, q-1\}$. Then, for all $[n] \in G$, we have

$$e^{\frac{2\pi i k n}{q}} = e^{\frac{2\pi i k' n}{q}}.$$

This implies that

$$e^{\frac{2\pi i(k-k')n}{q}} = 1 \quad \text{for all } n \in \mathbb{Z}.$$

In particular, for $n = 1$, we have

$$e^{\frac{2\pi i(k-k')}{q}} = 1.$$

This implies that $k - k' \equiv 0 \pmod{q}$. Since both k and k' are in the range 0 to $q - 1$, we must have $k = k'$. Thus, the characters χ_k are distinct. Therefore, we have found q distinct characters of G . By the previous corollary 3.4, we have $|\widehat{G}| \leq |G| = q$. Thus, we conclude that $|\widehat{G}| = q$.

For the $\widehat{G} \cong G$ part, we can define the map $\varphi : G \rightarrow \widehat{G}$ by $\varphi([k]) = \chi_k$. It is easy to check that φ is a group isomorphism.